

What is Vegetable and Difference between Fruits and Vegetables

Vegetable:

Any part of crop consumed as fresh or after cooking is called as "vegetable". Still definition is incomplete. There cannot be any single definition covering all characteristics.

Difference between Fruits and Vegetables:

Following are important differences between fruits and vegetables:

1. In case of fruits only flower or fruit is consumed While in vegetable all parts from roots to fruits are consumed (roots / leaves / stem / flower / fruit / seed / buds).
2. Fruits are consumed a fresh without cooking eg. Banana, apple, guava while vegetables are consumed a fresh as salad and also after cooking few are eaten both way fresh and after cooking eg. Tomato, onion, cucumber. etc.
3. Biennial Fruits are perennial and woody in nature with exception that some are annual and biennial and e.g. papaya, banana, while majority of vegetables are annual while few are biennial (onion, Cole crops, root crops) and few perennial (asparagus, parwal, tondali, hadga, drumstick).
4. Orchard management: Tranning, pruning and certain specific operations are performed in fruit crops and not in gegetable crops.
5. Fruit crops are mostly propagated by vegetative method except papaya, phalsa, Coconut and kagzi lime which are propagated by seed while majority of vegetable crops are propagated by seed with exception of tondali aspaaragus, sweet potato, parwal, potato.

Importance of Vegetables

1) Nutrition:

Vegetables are rich and comparatively cheaper source of vitamins. Consumption of these items provides taste, palatability, increases appetite and provides fiber for digestion and to prevent constipation. Their consumption in plenty fair amount of protein. They also play key role in neutralizing the acids produced during digestion of pretentious and fatty foods and also provide valuable roughages which help in movement of food in intestine.

Some of the vegetables are good sources of carbohydrates (leguminous vegetables, sweet potato, potato, onion, garlic and methi) proteins (peas, beams, leafy vegetables and garlic) vitamin A (carrot, tomato, drumstick, leafy vegetables), Vitamin B (peas, garlic and tomato), Vitamin C (green chillies, drumstick leaves, Cole crops, leafy vegetables and leaves of radish) minerals (leafy vegetables, drumstick pods). As per dietician, daily requirement of vegetables is 75 - 125 g of green leafy vegetables, 85 g of other vegetables and 85 g of roots and tubers with other food.

2) Importance as Food:

Food production is increasing. If is essential to sustain increased production besides nutritional standard of people. It can be increased by increasing production of vegetables which will help to solve food problem as yield of vegetable crops is 4 to 10 times more than cereals. Thus, vegetables play a vital role on food front as they are cheapest-sources of natural foods and can admirably supplement the main cereals of the country.

3) Importance to a grower:

Nature is in providing us-with all kinds of vegetable crops that can be grown in different seasons of the year in region. Different kinds of vegetables provide leaf, stem, flower, fruit or seed for consumption.' Considering vividness in the requirement of soil and season farmers can grow vegetable crops throughout the year for earning regular and steady income to meet the daily expenditure. There are vegetables of very short duration that can be grown as rained and intercrops in either agronomical crops or vegetable crops. There 'are vegetables which will improve soil and also provide fodder to catties. Thus farmer has wide choice to select suitable crop to adjust in his cropping pattern in given situation. Climate and soil conditions of this region are conducive to grow different vegetables.

4) Employment:

Since cultivation of vegetable crops involve intensive cultural operations starting from sowing to marketing, it provides more and regular employment opportunities in rural areas.

5) Industrial importance:

The perishable nature of vegetables demand comprehensive planning for movement, Storage, processing and distribution of vegetable products. The growth of vegetable industry as a commercial proposition largely depends on mainly allied enterprises like storage, processing marketing and maintenance and service enterprises to encourage vegetable growing.

The value of vegetables as an important article of daily human diet has come to be recognized all over the world in recent years. We get many specific chemical substances needed by our body for growth, reproduction and for maintenance of health. Vegetables contribute vitally to the general well-being due to the following reasons:

1. They are rich sources of 'Protective' elements like minerals, salts, vitamins and other chemical substances, which the human body needs to maintain good health and cheer.
2. Per acre yield of vegetables is very high.
3. They are an important source of farm income.
4. They have high aesthetic value.
5. More vegetable crops can be raised in one year.
6. Importance of vegetables in farmer's economy.
7. Vegetables are important .source of farm income:
 - a. Vegetables are sold at a higher rate than other crops. It provides regular as well as good source of income in addition to the income from the agronomic crops.
 - b. It provides regular work throughout the year to the farmers and his family labours.
 - c. Per acre yielded vegetables is very high: Vegetables give very high quantity of food per acre and they grow quickly. It is found that vegetables give higher yields in comparison to other crops.
 - d. More Vegetables can be raised in one year: Most vegetables are short duration crop and it as compared to other crops can be raised throughout the year Some of Vegetables (i.e. potato, brinjal, spinach, pumpkin, lady's finger etc.) can be grown twice and even three a year, some green vegetables become ready for harvesting within 15-60 days of sowing.

6) Importance of vegetables Production for medicinal properties:

Many of the vegetable crops possess high medical value for curing certain diseases. For instance, onion and garlic are found to possess antibacterial property (Sharma et. al. 1976). Many solanaceous and cucurbitaceous vegetables are found to possess Vitamin

Scope of Vegetable Growing in India

1) List of Indian Man:

One of the reasons for low agricultural production in India is "Low working capacity" of the majority of the population. At present the diet of an average Indian is ill balanced. It consists mainly of cereals. On an average we consume 375 gms of cereals, and only 30 gms of vegetables daily, as against 328 gms of cereals, 316 gms of vegetables- and 362 gms of fruits advanced countries. The dietitians recommend the consumption of at least 200 gms of leafy vegetables and 150 gms of root vegetable daily for a balanced diet. Thus to improve our diet, we should necessarily increase the vegetable production.

2) Area Under Vegetable Crops:

At present, greater than 70 % of our population is engaged in Agriculture over an area of 320 million acres. Out of this hardly about 1-2 percent of the total cultivated area is under vegetable crops. These figures showed the necessity of vegetable cultivation on larger area.

3) Higher yield from vegetable crops:

On an average, the yield of vegetable crop is about 5 to 10 times more than these of cereals. They are quick growing and shorter duration. Therefore it is time now, to take up the intensive and multiple vegetable cropping in India.

4) Present Agricultural policies:

Recently more attention is being devoted to increase the Agricultural production. This is being implemented through crash programmes, emergency and applied programmes, for this good seeds, irrigation facilities, knowledge of improved techniques of cultivation, proper plant protection measures quick transport, and better storage facilities are being made available through these programs.

5) As literacy increases in our country people realize the nutritional importance of vegetable in daily diet which results in increasing demand for vegetable in our country.

6) Due to construction of new irrigation projects lifting irrigation scheme & community well, irrigation facilities are increasing which help to increase area and production of vegetable.

7) Due to increase in fertilizer industries and their capacity of production, there is lot of scope for vegetable production.

8) National Seed Corporation of India, Indian Agricultural Research Institute. New Delhi, & Agricultural Universities of country-predicting & distributing pure seeds of improved varieties of different vegetable of all part of country.

9) Due to availability of cold storage, facility (in some part of country, it will be possible for cultivator of that part share his produce in cold storage after harvesting.

10) The auto-rick has also made it possible for a vegetable producer to take advantage of selling his produce to distant markets.

11) When there is large production of any vegetable the price rates in market come down for that vegetable. Naturally cultivator does not get more- income. But through preservation cultivator can get more price.

In this way there is vast scope for vegetable growing in India.

Statistics of Area and Production of Vegetable in India

Statistics on area and production of vegetable crops is not so accurate. It is highly essential to develop some procedure to work out proper figures of area and production. It is most important for planning its production as per population. As per recommendation of different experts every adult has to consume 250-300 g of vegetable per day.

India is the second largest producer of vegetable with 2.8 % of total cropped area under vegetables. An increase of 2.5 % per year in vegetable production is also necessary. Present production of 1.5 million tones of vegetable supply only 145 g per capita per day against recommended requirements of 300 g.

Area and Production of Different Vegetable Crops in India:

Sr. No	Vegetable	Area (000 ha)	Production (000MT)
1	Potato	1120	18500
2	Cabbage	220	4200
3	Tomato	330	5000
4	Cauliflower	280	500
5	Onion	405	4300
6	Brinjal	310	3300
7	Beans	145	390
8	Peas	148	2150
9	Cucumber	17	114

Area Production and Productivity of Different States:

Sr.No	States	Area (M ha)	Production (MT)	Productivity (t/ha)
1	Orissa	0.87	8.7	10.0
2	Uttar Pradesh	0.86	13.88 I	16.13 III
3	Bihar	0.85	12.28 II	14.44
4	West Bengal	0.51	5.39	10.56
5	Karnataka	0.29	5.70	19.65 II
6	Kerala	0.24	2.78	11.58
7	Maharashtra	0.21	2.95	14.04
8	Tamil Nadu	0.17	4.39	25.82 I

Source: NHB, 1998:

In India growing of vegetable is 4.8 times more remunerative than cereals and other field crops. Vegetable cultivation generates more employment. India has to find way to accelerate the vegetable production by developing high yielding varieties, varieties resistant to pests and diseases, hybrids and production technologies. Still there is a need to achieve target for supply of minimum 200 g and Opt 300 g. vegetable per capita per day.

Export of Vegetables

Sr. No	Fresh Vegetables	Qty (MT)	Value (Rs. Lakh)
1	Onion	427011	26521
2	Tomato	5300	-
3	Cucumber and gherkin	9608	1448
4	Mixed vegetable	3659	4385

Export of Processed Vegetable:

Sr. NO	Fresh Vegetable	Qty. (MT)	Value (Rs. Lakhs)
1	Mixed vegetables (Frozen)	6095	824
2	Dehydrated	4124	1781
3	Dehydrated garlic powder	744	246
4	Dehydrated garlic	233	87
5	Dried garlic	260	76
6	Other dehydrated vegetable	452	168
7	Preserved	15391	2690
8	Canned vegetable	1066	455
9	Other processed vegetable	5864	1440
10	Dried potato	899	103
	Total	88675	23107

Important Countries to Export Vegetable:

1. Gulf countries.
2. Adjoining countries viz. Pakistan, Bangladesh, Lanka, Nepal.
3. Europe, where nontraditional vegetables are exported.
4. South Asian countries and Australia. Onion and garlic are transported by ships and other vegetables by air.

Requirement of Variety and Quality of Vegetables:

1. The requirement of foreign market in terms of varieties and qualities differ.
2. In Europe and Japan, yellow/ brown, mild – pungent onion of big size are needed.
3. In south Asian countries light red to dark red strong pungent and varying sizes are needed.
4. Lanka and East Asian countries demand small, rose and multiplier onion.
5. Requirement of size, shape, colour, etc of different vegetable in various countries varies considerably.

Quality Parameters for Exporting Vegetable

Sr. No	Crop	Specific Requirements
1	Okra	Green tender, 6 -9 cm long. In Europe 5 -7 cm long.
2	Chili	Green 6- 7 cm long.
3	Bitter gourd	Green, 20- 25 cm long short neck.
4	Bottle gourd	Light green, straight cylindrical, 25-30 cm long.
5	Gherkin	Green small sized having 160- 300 fruit/kg.
6	Tomato	Round, medium, red colour, in middle east and cherry tomatoes in Europe.
7	Big Onion	4-6 cm light to dark red colour, round shape, strong pungency in gulf markets and south east Asia, 3-4 cm light red and round shape in Bangladesh Yellow/ Brown colour 7 -8 cm round or spindle shaped in Europe and Japan.
8	Small Onion	Dark red, 2-3 cm and round.
9	Garlic	White, round, 5 cm and above bigger coves of 10- 12 mm and 10- 15 size bulbs accepted.
10	Potato	White, oval, 4.5 to 6 cm size, Bangladesh red types, Iraq-Iran yellow fleshed.

Export Policy and Period Required to Export

1. Fresh vegetable export other than onion is allowed freely. Under this category and person may export vegetable to any country where export any person may export vegetables to any country where export is prohibited.
2. Government has started giving cash incentive of 10- 12 %.
3. Export of onion up to 20 kg per consignment is allowed by air as assorted vegetables.
4. Onion export has been canalized through (NAFED).

Period of Requirement:

1. South East- Asian countries demand is all year round. It slightly differs with local production.
2. In European market demand is from Nov. to Dec and April to May when there is no local production.
3. India having varied agro climatic condition can produce and supply the vegetables to different markets from one part or other parts of the country round the year. If necessary infrastructure facilities are created for pre- cooling and transport.

Problem in Export of Vegetables

1. Inadequate air cargo space.
2. Higher air freight.
3. Restriction in the export. During shortage of crop export is restricted. Thus regular export is stopped.

4. Non- availability suitable variety.
5. Absence of cultivation of suitable crop varieties exclusively for export is fresh for processing.
6. Improper packing of the produce.
7. Vegetables are packed in jute (undesirable).
8. Corrugated fiberboard boxes are not of enough strength and damaged in transport.
9. Cold storage facilities at all air ports.
10. Inadequate handling space.
11. Inadequate research development back up.
12. No export oriented research programmes are being taken up in vegetables.
13. Quality of vegetables is not uniform.
14. Cost of vegetable washed in sorting and grading.

Suggestion for improvement in Export of Vegetable

There is need for such policy so that quality crops at competitive price are available regularly in market.

Following are Suggestion:

1. Long term and consistent export policy, Avoid frequent change in export policy.
2. Exclusive production of different vegetable for different season in suitable packets be arranged to have regular supply throughout the year at competitive price without affecting domestic supply.
3. Enough quality control measures should be made.
4. Regular assessment for international market.
5. Suitable diversification for export both in term of countries and commodities for increasing quality and value.
6. Identify suitable pocket for growing different vegetables.
7. Develop agro technique and for integrated nutrient, P and D management, organic farming etc.
8. Production and distribution of quality seed in adequate quantities be arranged.
9. Pre-cooling and cold storage facilities are increased.
10. Ventilated and temperature controlled, storage and transport facilities are necessary.
11. Packing material.
12. Export promotion programme be carried out in other countries to prove superiority of taste, flavor of Indian Produce.

Vegetable Garden

Vegetable gardening may broadly be divided into four parts on the disposition that is to be made of the products.

1. Home or Kitchen or Amateur Gardening:

In which the purpose is to raise a supply for the use of family only.

2. Commercial Vegetable Gardening:

This is the method of growing vegetables on large scale for the sale according to the objects sought and the methods employed in producing and disposing of the crops.

This Further Divided into:

- 1) Market gardening.
- 2) Truck gardening or farming.
- 3) Vegetable forcing.
- 4) Vegetable for processing:
 - i) Canning
 - ii) Freezing
 - iii) Dehydration
 - iv) Pickling & fermentation
 - v) Vegetable gardening for seed production
 - vi) Floating- Veg. garden.

Home or Kitchen Gardening

The main purpose of the kitchen or home vegetable gardening is to provide the family daily with fresh vegetables rich in nutrients and energy. There is to be a more or less continuous supply of vegetables throughout the year according to the season. In kitchen gardening emphasis is usually placed on a great variety of crops according to season. Home gardening idea is spreading among more and more housewives due to the increasing awareness among them that just as good placement of shrubs and flowers adds to the beauty of the house-Kinds of Kitchen gardening:

- 1) A combined flower and vegetable garden.
- 2) Exclusive vegetable gardens But these are rather less common at least in houses in urban areas. They also vary according to the size of the residential plot, etc.
 - a) In large size plot almost every important vegetable can be grown in them due to enough space available in the back yard.
 - b) In medium size plots the area is not enough so that even an assorted collection of vegetables cannot be accommodated.
 - c) In very small houses - the area available for kitchen garden is also very small, so only a few small beds of vegetables, supplemented with narrow strips here and there are available along the boundary wall, and other locations.
 - d) Terrace gardening or growing vegetables in pots- In cities and towns where there is little garden space available one can grow vegetables in pots drums and other container; placed on terrace or roof.

Advantages of kitchen gardening:

- 1) Kitchen gardening is the best means of recreation and exercise. It is an excellent hobby and a healthy occupation in spare time for the young and the aged. According to Venkataratnum (1963) "An hour or two spent either in the morning or evening in kitchen garden provide good exercise to the body and a healthy recreation to the mind.
- 2) Kitchen gardening helps in lowering down the vegetable bill. There are no transport charges and middlemen's share which greatly add to the price paid by the consumers in purchasing vegetables from the market.
- 3) According to Dr. Mehta (1959), "It is an ideal medium for training children, through gardens, in beauty and order".
- 4) Kitchen gardening secures enough vegetables within the means of all classes at a very cheap rate. It is often remarked that it is cheaper to buy vegetables than to raise them.
- 5) The cost of raising vegetables in the kitchen gardening through ones owns labour is far less than what a family spends on vegetables in the market.
- 6) Vegetables grown, in the kitchen gardens are fresh and are not liable to infection with germs occurring in unsanitary markets. Thus they can be used freely without any trouble.

Market Gardening

It is that branch of vegetable growing whose object is to produce vegetables for the local market. It is one of the most intensive types of vegetable gardening where the most skilful methods for growing of vegetables for commercial purpose are employed. It is developing day by day according to the rise in population for meeting the needs of those persons who have no land. Formerly the market gardens were located near the cities within 16-24 kilometers radius but now-a-days, with the building good roads and the development of automobiles, the area has been extended considerably. Now vegetables are brought to mandates from distant village by trucks.

Due to transport facilities, markets are no longer local. Consumers can get vegetables even from a distance of 500 kilometers in off season. Now-a-days market gardening probably, supplies almost all the vegetables. Timely and proper irrigation facilities good seed, enough fertilizers and manures, abundant labor supply and easy transportation facilities are essential for market gardening. For market gardening crops like cauliflower, tomato, onion, pumpkin and other such vegetables are planted,. Market gardening is developing these days and there is a possibility of specialization in production of vegetables in the regions especially suited to the production some particular vegetables.

Truck Gardening or Farming

It is the method of growing special vegetable crops in relatively large quantities for distant markets by the employment of intensive methods in this method of gardening there is. Comparatively less intensive cultivation and land values are also lower than market gardening. The development of national highways in India and efficient motor truck facilities is recent years has helped his considerably. Truck farming becomes prosperous in those regions where climate, soil and other factors of production assure better yields than the average, and where labour is cheap. Through truck fanning vegetable surplus in one region can be better disposed of in those areas where there is a shortage of vegetables. Onion from Nasik and green chills from UP. are available in Gwalior during early summer season due to this .Better storage facilities are not available, everywhere in India and this affects considerably the truck gardening.

Vegetable Forcing and Vegetable for Processing

Vegetable Forcing:

It is the method of growing of vegetables out of their normal in sellers, heated buildings, greenhouses, cold farms and under other artificial growing conditions. It is the most intensive type of vegetable growing. Vegetable produced through these, give a very high return, vegetable forcing is not possible because the consumers cannot afford the prices of such vegetables. According to Choudhury (1997) growing of the summer vegetables on the river beds during the winter months with the help of organic manure, with breaks of dry grass is also type of forcing. Some times for early product seedlings of crops like tomato or brinjal are forced to germinate in small protected structures. This may also be a type of vegetable forcing.

Vegetable for Processing:

Low cost with high quality and convenience in storage and use, have given rise to vegetable processing in India. But this industry is still not well-developed. People also in general, have not yet developed liking for the canned and processed vegetables in India. Still we need vegetables of good quality for canning, dehydration and freezing in our existing factories. Pickling and fermentation also require vegetables. For these also we must grow good vegetables. Vegetables for processing are generally grown around vegetables processing factories, for the regular supply of vegetables to the factories.

vegetable Gardening for Seed Production and Floating Vegetable Garden

Vegetable Gardening for Seed Production:

As the vegetable cultivation is increasing day by day need for good vegetable seed has also increased. To meet this demand State Government Department of Organizations like National seeds Corporation of India are producing registered and certified seeds at their own farms or on cultivators fields. Though this is a new and expanding industry in India but it has a bright future.

Floating Vegetable Garden:

This is also a type of vegetable gardening generally seen in Dal and other lakes of Kashmir Valley. For this type of garden first of all a floating base is made from the roots of various grasses and weeds growing wild in the lake. After this leaf compost and wild vegetation's of the lake are spread over them and then seedlings are transplants. Irrigation and other subsequent operations are done by with the help of boats. Seasonal flower are also grown in these gardens.

Classification of Vegetable Crops

There are many numbers of Vegetables. Different parts are used for consumption. They are botanically different and their climatic and cultural requirements are different.

There are four Main Methods of Classification Based on:

1. Education Botanical relation
2. Based on Hardiness (Tolerance to low temp)
3. Parts Used for consumption
4. Methods of Culture

1. Botanical Classification:

This method of classification is based on botanical relationships of crops in responding cytology, morphology, taxonomy and cross ability. This is useful to be a guide for crop improvement and seed producer for deciding isolation distance. The cultural requirements may vary ex. Solanaceous family includes potato, chili and brinjal, There are same cultural requirements. At the same time cucurbitaceous have similar cultural requirements and common pests and diseases. Botanical name avoids confusion in name as common names are different but scientific names are common all over world. It gives information on class, family, genus, species, variety etc Ex. Solanaceous

family, Potato, brinjal, chili, cucurbitaceous family: Melons, gourds. Not useful to grower. Thus botanical classification is useful to breeder, seed producer and to avoid confusion in common name.

2. Classification Based on Hardiness:

Vegetables are grouped as hardy or tender on the basis of tolerance to frost, tolerance to lower temperature. This gives information of season of growing i.e. summer or winter. Not useful to grower, since soil and climatic requirements are

Hardy	Semi Hardy	Tender
Asparagus	Beet Root	Okra
Cabbage	Carrot	Brinjal
Garlic	Cauliflower	Tomato / Chili
Onion	Palak	Beans
Peas	Potato	Cucurbit
Radish		Sweet Potato
Spinach		Sweet Potato
Spinach		Amaranthus
Turmp		

3) Based on Parts Used for Consumption:

From roots to fruits different parts of vegetables are consumed. On that basis vegetables are classified.

Leafy Vegetable	Palak, Amaranthus Methi (leaves)
Root	Carrot, Radish, Turnip, beet root
Fruit	Tomato, Biinjal; Cucurbit
Bulb	Onion, Garlic
Tuber	Potato, Sweet Potato

4) Based on Method of Culture:

In this method all those crops having similar cultural requirements are grouped together. They" are botanically different. System has practical utility for vegetable grower. In this method one can generalise cultivation practices for one group and thus avoid repetition individually for all crops. There are 11 classes. In some cases they are botanically also similar Ex. cucurbits. On excises for convenience they are grouped as under.

I.	Perennial Vegetable	Asparagus, cocinia (Tondali), Parwal, Drumstick
2	Greens	Spinach
3	Salad Crops	Lettuce, celery
4	Cole Crops	Cabbage, Cauliflower
5	Bulb Crops	Onion, Garlic
6	Root Crops	Carrot / radish, turnip
7	Tuber	a) Potato, b) Sweet Potato
8.	Peas & Beans	Cluster bean, cowpea, Dolichas
9	Solanaceous	Tomato, Brinjal, Chili,
10	Cucurbits	Watermelon, pumpkin, gourds, Cucumber etc.
11	Okra	Okra

Other Basis of Classification of Vegetables

A) Thermo Classification of Vegetables:

Vegetables are classified as cool and warm season crops. Depending upon temperature requirement, cool season crops require cool weather,

i) Cool Season: They are characterized by :

- a. In all cases vegetative parts are edible ex. Leaves (Palak), roots (carrot). stem (kale) except peas. Fibrous root system.
- b. Need constant supply of moisture.
- c. Careful application of nutrients. Ex. Onion, garlic, Cole crops, root crops, Palak, potato, peas.

ii) Warm Season Crops: They are summer vegetables and characterized by:

- i) In this group reproductive parts of vegetable are used for consumption example. Tomato, Brinjal, Cucurbits and beans. (Except - Sweet Potato).
- ii) Tap root system.
- iii) Water requirement is less.
- iv) Tolerate more Temperature.Ex. Solanaceou8s except potato, sweet potato, okra, beans.

B) Miscellaneous Classification:

- a. Direct sown: Okra, Beams
- b. Transplanted: Brinjal, Tomato Onion, Cole crops
- c. By cuttings only:Tondali, Sweet Potato
- d. By bits of tubers : Potato, Yam

C) According to Lime Requirement:

- a. Require no Lime : Potato, Tomato, and Radishb.
- b. Require lime medium : Pumpkin, Pea, and Cauliflower
- c. Require more lime : Brinjal, Cabbage, and Cauliflower
- d. Need much lime : Onion, Chilies

D) According to Forcing:

- a. Cool forcing : Onion, Pea, cauliflower
- b. Warm forcing : Cucumber, brinjal, Tomato

E) Rate of Respiration:

- a. Very high rate of respiration : Pea, spinach
- b. High rate of Respiration : Beans
- c. Moderate rate of respiration : Carrot, Tomato, Chilli
- d. Low and very low : Cabbage, Onion, and Potato.

F) Tolerance to Soil Reaction:

- a. Slightly Tolerant (pH 6.8.6) : Cauliflower, Palak, Musk melon,
- b) Moderately Tolerant (pH 6.8.6): Bean, Carrot, Garlic, Tomato,
- c) Highly Tolerant (pH 6.8 - 5-5): Potato, Sweet Potato, Watermelon.

G) Classification Based on Salt Tolerance:

- a) Semi tolerant 0.25 : Tomato,
- b) Medium Tolerant 0.50 - 0.75 : Chilies, Okra, Cabbage, Sweet potato, Onion,
- c) Highly tolerant 0.1 - 1.20 : Ridge gourd, Bitter gourd However, no one method satisfies all groups of workers.

Cultivation of Tomato (*Lycopersicon esculentus* Mill)

Botanical Name:

Lycopersicon esculentus Mill

Family:

Solanaceous

Botany:

The tomato Fruit is a fleshy berry.

Package of Practices of Tomato Cultivation as Follows:

1. Climate and Soil Requirement
2. Sowing Time and Seed Rate
3. Layout and Spacing
4. Manures and Fertilizers
5. Improved Varieties
6. Irrigation
7. Intercultural operations, Pruning and Staking / Training

8. Harvesting and Yield

9. Insect Pest Control

10. Diseases Management

Climate and Soil Requirement for Tomato Cultivation

Climate:

Tomato is warm season crop. Grows well in those regions that are free from frost. It can't be grown successfully in places of higher rainfall. Temperature after tomato crops in following ways.

1. Optimum temperature for seed germination is 26 to 32°C.
2. The optimum temperature required for its cultivation is 15 - 27°C. At higher temperature its blossoms drop off. The damage is great when high temp is combined with dry wind. It will result in the failure of fruit set due to drying of stigmatic liquid.
3. Colour development: In tomato red colour is due to the pigment Lycopene. Lycopene is highest at 18 to 26°C while production of this pigment drops off rapidly above 30°C and 'nil' above 40°C.
4. Carotene is developed rapidly at high temperature.
5. If fruits exposed to direct sunlight, their tops may turn whitish yellow & become leathery in texture. This is common in late varieties during summer season. This condition is known as sun - scald.
6. A warm, sunny weather is most suited for proper ripening, colour, quality & high yield.

Soil:

Sandy loam soil with a well drained clay sub soil is best suited. Light soils are good for early variety. While clay loam or silt loam soils are well suited for heavy yield (Late variety), grows at pH 6.0 to 7.0 satisfactorily. The soil should be well prepared & leveled by ploughing the land 4 - 5 times.

Sowing Time and Seed Rate for Tomato Cultivation

Sowing Time and Seed Rate:

In our region tomato can be grown throughout the year. Sowing is done in May - June for kharif crop, August - Sept for Rabi Crop and December - January for summer crop. The seeds are sown in well prepared nursery beds 500 gms to 600 gms seed is enough for planting one ha of land. When the plants are about 3 to 4 weeks old, they should be transplanted.

Layout and Spacing for Tomato Cultivation

Layout and spacing :

Ridges and furrow type of layout is used. The spacing recommended for tomato crop is 60cm to 75 cm x 60 cm.

Manures and Fertilizers for Tomato Cultivation

Manures and Fertilizers:

30 to 50 cartloads of FYM or compost should be applied in the soil & incorporated in it during field preparation. 75 to 100 kg P and 50kg K should be applied for getting better yield of tomato.

Improved Varieties of Tomato

Improved Varieties:

Marglobe, Pusa Rubi, Roma, Sioux, Pusa early dwarf, Punjab chohara. Devgiri, ATV-1.

1. Bhagyashree:

Released during 1992 especially for processing purpose. Fruits are medium to big (70- 80 g) and round in shape, dark red in colour, high lycopene content and low seed content Average yield is 520 quintals per hectare. (M.P.K.V. Rahuri).

2. Dhasashree:

Developed from a cross between 407D3P2DBK x LA-124 by employing pedigree method of selection and released in 1992. Fruits medium small (60-70 g) and round with orange colour. Tolerant to spotted wilt virus and leaf curl virus. It has high yield potential (600-800 q/ha). Suited for local fresh markets. (M.P.K.V. Rahuri).

3. Rajashree:

It is F1 hybrid developed for distant marketing and released in 1994. Plants are semi-indeterminate. Fruits are of medium size (60-80 g) with orange red colour having good keeping quality. It is less susceptible to TLCV. Average yield is 750 quintals per hectare. (M.P.K.V. Rahuri).

4. ATV 1:

It is a determinate type of variety. Leaves are large with dark green colour. Fruits are medium size (90-95 g/fruit) pear shape with blood red colour, pulp colour is rosy and green. It is moderately resistant to early blight. It produces 400 - 500 quintals fruits in a hectare. It is released in 1985. (M.A.U. Parbhani) .

5. Devgiri:

Davgiri (ATV-2) is superior to ATV-1 and Pusa Ruby. Fruits are of medium size (90-95 g/fruit). Fruits have good taste and attractive colour. It is resistant to early blight and leaf curl virus diseases. Average yield is 550 quintals per hectare. (M.A.U. Parbhani).

6. ATH-1:

It is F1 hybrid between male sterile female ACC 2549 and male parent Hawai 7992. Average yield is 550 quintals per hectare. It has indeterminate growth habit.

Average fruit weight is 110-150 g TSS is 7.0 per cent and fruit colour is attractive red. It is resistant to early blight disease. (M.A.U. Parbhani).

7 .Vasimdhara:

It is the FI hybrid between sterile female .PMSK -2 and male parent Kalyanpur T-2, recommended in 2000. Determinate type, fruit weight 70-75 g and colour is attractive red. Resistant to early blight and spotted wilt virus disease. Average fruit yield is 550 - 600 quintals per hectare.(MA.U. Parbhani).

8. Parbhani Yashisree:

It is recommended in 1999. It is a hybrid derivative of the interspecific cross between *Lycopersicon esculentus* x *L. hirsutum* and back cross to ATV-2. It is resistant to early blight and produces 13 per cent higher yield than Bhagyashree and ATV-2. Average yield is 600 quintals per hectare. (M.A.U. Parbhani).

9. Sonali:

It is a high yielding, good quality and bacterial wilt resistant variety. Sonali was released during the year 1986. It yields about 200-250 quintals per hectare, suitable for processing. It has good keeping and transport qualities. (BSKKV, Dapoli)

Irrigation for Tomato Cultivation

Irrigation:

It should be so arranged that the soil remains continuously moderately moist. Avoid excessive irrigation as it induces the plants to run to vines and* drop the blossoms off There is no need of irrigation during rainy season if, there is a proper distribution of rainfall. When the rainy season is over, the crop may be irrigated twice or thrice in a month. The crop planted during the winter season will need .irrigation once in about 20 day. While the crop transplanted during spring season will need irrigation more frequently, say, one in ten days in the beginning, and later on during the hot season, the tomato crop is irrigated once a week. During the winter season, whenever there is a danger of frost, the crop must be irrigated so that the temperature may not go down too low and damage the plants. Tomatoes, that have been growing when moisture is low, may split severely after a rain of even after a heavy-Jew to an influx of water into the fruit through roots, stems or through the cracks in the skin near the calyx. Therefore, the crop should be irrigated carefully during the fruit ripening stage.

Intercultural Operations, Pruning and Staking/Training in Tomato Cultivation

Frequent inter-tillage and cultivation should be done in the field to keep it free from weeds. As the plants grow bigger, all the intercultural operations should be shallow, so that the roots which spread unto a depth of 5 cm below the surface soil may not be injured.

Though pruning and staking in the tomato .crop increase cost of cultivation, yet according to Mehta (lit. cit.) and Fernandez (1963), both these operations make a tomato-plant grow more better and larger tomatoes resulting in a higher yield per hectare. But both of these operations depend on the time that is at the disposal of the grower and the availability and cost of the staking material.

According to Fernandez, through pruning tomato plant may be made single-stem in the following ways:

All the side shoots are removed so that plant may use all its food and energy to develop its fruits and very little of it is left for the foliage. These are then staked. In this way the plants get a 1.2 meter square area and bear more fruits.

The following are the advantages of pruning to single stem and staking:

- 1) They produce more fruits per hectare per plant an average.
- 2) They yield earlier and give rise to more cluster.
- 3) They make picking more rapid .and cultivation and spraying easier.
- 4) They give cleaner more uniform and lager fruits.

When the fruits ripen, the plots become a treat to the eye with their ripe red colour. According to Fernandez, the vine ripened tomatoes from single -stem plants are as nutritious and tasty as the fresh laid eggs.

Disadvantages of this method:

- i) Cost of cultivation increases considerably.
- ii) According to Mehta, there are more changes of sun scalding in the excessively warm weather due lo fruits being exposed to sun.

Harvesting and Yield of Tomato Cultivation

Harvesting:

Tomato fruits are picked up from the plants by grasping them by the hand and dislodging them from the vine by twisting keeping the thumb pressed against the vine.

According to the Use of Fruits, they are Harvested in Following Stages:

1) Green Stage:

About a fortnight before turning (development of a trace of redness at (he styler end of the fruit), the fruits will develop normal colour of the vine though they are still green yet they may be fully developed. These fruits are picked and sent to distant markets.

2) Pink Stage:

At this stage red or pink colour on the fruits varies from a trace at the blossom end to a considerable extent covering the surface. Though at this stage most of the fruits are red, yet they are not fully ripe. They are picked for local markets.

3) Ripe Stage:

At this stage the surface of most of the fruits is red and the softening of the fruits begins. They may be picked for home or table use.

4) Full Ripe Stage:

At this stage the fruits have approached maximum colour development and may feel soft to the touch. Now the starch is changed into sugar. They are ordinarily used within 24 hours of picking and are consumed or used for canning and pickling.

Yield:

It varies 160 to 200 quintals per hectare according to season and variety planted Duration of crop ranges from 160 to 180 days.

Insect Pest Control in Tomato Cultivation

Insect Pest:

1. Fruit Borer:

Damages leaves shoot & fruits. Fruits become unfit for consumption.

Control:

Spray carbaryl 50 WP @ 0.3% or Carbaryl Dust 10% @ 35kg/ha. Infested shoot & fruit removed & destroyed.

2. Jassids:

They suck the sap from tender part & leaves.

Control:

Spray Endosulfan 35EC @ 0.5% or Phosphomidon 85 @ 0.2%.

3. Mealy Bug:

A scale insect that covers up the whole plant and suck the sap from the plant

Control:

Removal of infected shoots and spraying the plant with Malathion (Cythion 50 EC, Malathion-50 EC) @ 1ml per litre of water is the best method of controlling this pest

Diseases Management in Tomato Cultivation

Diseases:

1) Early blight:

Dark brown to black spots are formed on leaves & stem with concentric rings on them. Plant growth is affected, fruit size & yield is also affected. Control measures : Spray copper Oxy-chloride 50 WP @ 0.30% or Zineb 75 WP 0.2%.

2) Damping Off:

Affected seedling rots at collar region in nursery bed due to stagnation of water in nursery.

Control measures: Treat seed with 1% Mercuric fungicide @ 2.5—3.0 gm/kg. of seed before sowing.

3) Fruit rot:

Brownish spot appears on fruit at the point of contact between fruit and soil. Fruit decays rapidly and becomes unfit for consumption.

Control Measures:

- i. Support the plant properly, plant on ridges in irrigated soil.
- ii. Don't plant on poorly drained soil,
- iii. Spray Bordeaux mixture.

4) Virus diseases:

According to Chauhan (1967), tomato mosaic is the most common virus disease. It causes light and dark green mottling hi the foliage followed by slight curling and some malformation of the leaflets. The plants become some what stunted in the beginning. and later on-fruit setting is also reduced. According to Mehta (1959), various strains of this viruses striking yellow mottling of leaves, stems and fruits with curling dwarfing and distoration of the foliage and also dark brown streaking of the stems and leaves.

Control measures:

- i. Uprooting and burning the disease affected plants as soon as they are noticed is the best method to check the spread of this disease.
- ii. The insect vector should be controlled by spraying the crop with Dimetnoate (Rogor 30 EC) or Oxy dernition methyl (Metasystox-25EC) @ I ml or phosfamidon (Dernicron-1 OOE) @ 0.5 ml per litter of water.

Cultivation of Brinjal or Eggplant (Solanum mdongena)

Botanical Name:

Solanum mdongena L.

Family:

Solanaceous,

Package of Practices of Brinjal Cultivation as Follows:

1. Climate and Soil Requirement
2. Sowing Time and Seed Rate
3. Layout and Spacing
4. Manures and Fertilizers
5. Improved Varieties
6. Irrigation
7. Intercultural operations, Pruning and Staking / Training
8. Harvesting and Yield
9. Insect Pest Control
10. Diseases Management

Climate and Soil Requirement for Brinjal Cultivation

Climate:

1. It is a warm season crop and is very susceptible to frost.
2. Late round varieties are more tolerant of frost than the early long varieties. Cool nights and short summers are unfavorable to its satisfactory yields.
3. It requires a long growing season with the high average day and night temperatures. Its seed germinates well at 77 °F temperature.
4. A daily mean temperature of 13-21°C is most favorable for successful production.

Soil and its Preparation:

1. Though it can be grown on different types of soils, yet it is grown with considerable success in fine and rich loam soils that are deep and well-drained.
2. The soil pH should not be higher than pH 5.5 to 6.0 for its better growth and development. As the crop remains in the field for a number of months therefore, the soils should be well prepared by being ploughed 4-5 times before transplanting the seedlings. When the field is well prepared and leveled, the beds of suitable size are made in the field before transplanting.

Sowing Time and Seed Rate for Brinjal Cultivation

Sowing Time and Seed Rate:

Brinjal can grow twice during the year. The sowing time is May-June, August- September and December- January, The 700 gms to 1000 gms seed required to raise the seedlings for 1 ha, area.

Layout and Spacing for Brinjal Cultivation

Layout and Spacing:

Ridges and furrow type of layout is use. Seedlings are raised on raised bed. Spacing is 75 x 60 cm. to 75x75 cm.

Manures and Fertilizers for Brinjal Cultivation

Manures & Fertilizers:

30 to 50 cartloads of FYM or compost is applied at the time of field preparation. In brinjal 60 kg N, 50kg P. should be-applied per hectare.

Improved Varieties of Brinjal

Improved Varieties:

Manjari gota, Pusa purple round, Pusa purple long, Pusa purple Cluster, Pusa kranti, Pragati, Aruna, ABV-1, ABH-1. Local: Dorala, Bhatai, Hingana Dorala, Gulabi Dorala.

Brinjal (*Solanum melongena*):

1. Krishna:

Krishna is a Fuji hybrid released in 1991 by MPKV Rahuri for Western Maharashtra. Plants tall, hardy and resistant to lodging. Egg-shaped, spiny, purple coloured fruits with white stripes borne mostly singly or occasionally in pairs. Good colour retention. Good acceptance in the home market. Average yield is 480 quintals per hectare. (MPKV, Rahuri).

2. Manjari Gota:

Developed by selection from a local germplasm collected from village Manjari near Pune and released by the Department of Agriculture, in 1965 for Western Maharashtra, Plants are medium tall and spreading, spines on leaves, midribs and peduncle the fruits. Fruits are round purple -with white stripes. Average yield is 250 quintals per hectare. (MP.K.V. Rahuri).

3. Pragati:

Developed by selection from a cross Vaishali x Manjari Gota and released in 1988. It is an improved version of Valsbali. Plants are hardy and resistant to lodging, egg shaped purple fruits with-white stripes borne in clusters with spines on pedicel. Duration is 180[^]-190 days. Average yield is 350 quintals per hectare. (M.P.K.V, Rahuri)

4. Vaishali:

Developed by selection from a cross Manjari Gota x Arka Kusumakar and released in 1985. Plants are medium and spreading. Fruits oblong purple with 1 white stripes. Fruits are borne in clusters (3"-4"). Average yield is 350 quintals per hectare. (MPKV, Rahuri).

5. Anuradha:

It gives about 10 per cent higher yield than Manjari Gota. It has shining and attractive fruit colour (purple and white stripes). Fruits are borne in cluster with spines on fruits, leaves and stem. (MAU, Parbhani).

6. ABV-I:

This variety is released by MAU, Parbhani in 1985. It has become popular with the cultivators of Marathwada. It has erect plant habits with faint purple stem colour and dark green leaves. Fruits are small round (50 g/fruit) with green purple stripes and spines on the calyx. Average yield is 250 quintals per hectare. (MAU, Parbhani)

7. Aruna:

Notified during the year 1985-86. Plants are medium in height; fruits are oval in shape and violet in colour. Days to first harvest are 90 - 95. Yield is 300-385 quintals per hectare during kharif season. 200-225 quintals per hectare during summer season. (PKV, Akola).

8. PhuleHarit:

Developed by employing pure line selection. It is good for kharif season. Average weight is 190 g. Fruit length is 16.2-cm. Fruit colour is green with white stripes at the tip, fruit surface is smooth. It produces 25 fruits per plant. Average yield is 330 quintals per hectare. Number of pickings are 18. Duration is 180 days. (MPKV Rahuri).

Irrigation for Brinjal Cultivation

Irrigation:

Irrigate the field after every third or fourth day during the summer season and after 12 to 15 days during the winter season. Timely irrigation is very important for high yields of brinjal. Brinjal fields should be regularly irrigated to keep the soil moist during frosty days.

Intercultural Operations in Brinjal Cultivation

Intercultural Operations:

Inter-tillage or hoeing with some hand drawn implement should be done in the crop (in such a way that a root system is not damaged) to check the weeds. These operations should be fairly deep and close when plants are very small, but they should be made shallow as plants develop. The most serious weed in brinjal and other solanaceous crops is *qrobancbai* sp. It is a root parasite and should be controlled carefully according to the measures suggested by the experts.

According to Choudhury (1967), fruit set can be increased with plant regulators. These chemicals are either used for seed treatment or sprayed over the plants. Para-chlorophenoxyacetic acids, 2, 4, dichlorophenoxyacetic acid and naphthaleneacetic acid, have been found effective for this. Out of these chemicals the cheapest and easily available is 2,4-D. The seed can be soaked for 24 hours in a solution of 2, 4-D at 4 to 5 ppm before sowing or spraying the whole plant with 2, 4-D solution at 2 ppm when the first few flower clusters appear. These treatments give about 50 per cent earlier and higher fruit set.

Harvesting and Yield in Brinjal Cultivation

Harvesting:

Fruits are harvested when they are immature. They should be severed from the plant by cutting with small shears or a knife. Fruits are allowed to attain a good size and colour till they do not lose their bright, glossy appearance and become dull.

Yield:

In brinjal 250 to 300 quintals / ha.

Insect Pest Control in Brinjal Cultivation

Insect Pests of Brinjal:

1) Brinjal Fruit and Shoot Borer:

It is one of the major and serious insect pests of brinjal. A short pinkish caterpillar bores into the terminal shoot and eats internal tissue; it bores into the young fruit through the calyx, leaving on visible signs of infestation. The large holes usually seen on the fruits are the exit holes of caterpillar. The insect affected fruits become unfit for consumption.

Control Measures:

i) The insect affected part should be clipped along with insect and destroyed any fruit with holes should be picked and destroyed.

ii) The affected crops should be sprayed with phosphamidon (demeton-100 EC) @0.5 ml or diazinon (Diazinon-50 EC) @1 ml per litre of water at fortnightly interval or spraying with carbaryl 0.20% or Monocrotophos 0.05%

2) Brinjal Stem Borer:

A pale white caterpillar bore into the stem and kills the plant Control measures: Same as for brinjal fruit and shoot borer.

3) Leaf Eating Beetle:

The beetle and grub feed on the leaves and other tender parts leading to a considerable reduction in the yield of the egg plant.

Control Measures:

i. Hand picking of egg and larval is the best method of, controlling this pest if infestation is only in few plants.

ii. This insect can effectively be controlled by spraying crop with Endosulphan

(Thiodan -35 EC) or Phenthothion (Sumithion-50 EC)@ 2 ml or Fenthion

(Lebacid-1000 EC) or Thiomiton (Ekaton-25 EC)@ 1 ml per litre of water.

4) Nematode:

The nature of damage and control measures of nematode is same as for potato nematode. The egg plants are also affected by mite, Jassids, Aphids and Mealy bug. They suck the sap from the leaves and in severe cases the whole plantation looks yellowish and leaves drop down prematurely. These insects except mites can effectively be controlled by spraying the crops with Methyl parathion (Metacid 50 EC) or Oxidemeton methyl (Metasystox 25 EV) @ 1 ml or Malathion-50 EC @ 2ml per litre of water at fortnightly. Interval, The mites can be controlled by spraying the crop with Dicophol (Keithane-18.5EC) @ 2 ml or Morocite-40 EV @ 1 ml per litre of water.

Diseases Management in Brinjal Cultivation

Diseases of Brinjal:

1) Damping Off:

It is a serious disease of brinjal seedlings and mainly occurs in nursery bed. The disease infected seedlings rot at ground level and then the plants fall over ground. The seedlings die in patches.

Control Measures:

1. The seed bed should be treated with Formalin before sowing of seeds.

2. The seeds should be treated without water (30 minutes at 52°C) or Cerasan or Agrosan G.N. before sowing of seed.

3. The seedlings in the nursery should be sprayed with any fungicides at a regular interval.

2) Phomopsis Blight and Fruit Rot:

It is a serious disease of brinjal. The fungus attacks all parts of the plants above ground. Dark brown lesions appear on the stem and round to oval spots are formed on the leaves. Disease fruits show short and watery lesions which later on become black and mummified.

Control Measures:

- i. Use of disease free seeds, seeds treatment with some fungicide and long crop rotation are the most common remedial measures of this disease.
- ii. Disease resistant variety (Such as Pusa Bhairab, Pusa Cluster etc.) should be Cultivated,
- iii. The disease can effectively be controlled by weekly spraying of nursery and field with Zineb (Dithane-Z-78) or Mancozeb (Dithane M-45) @ 2.5 gm per litre of water.

3) Little Leaf of Brinjal:

The affected plant produces numerous tiny Yellow leaves and does not bear fruits. The disease is transmitted by leaf hopper.

Control Measures:

- i. The disease affected plants should be destroyed,
- ii. The insect vector should be controlled by spraying the crop with Dimethoate (Rogor-30 EC or Oxydemeton methyl (Metasystox-25 EG) or Monocrotophos (Monocil)@ 1 ml per litre of water to check the spread of this disease,
- iii. Disease resistant variety such as Pusa purple Cluster should be cultivated

Cultivation of Chilli or Capsicum (*Capsicum Annum* or *Capsicum frutescens*)

Botanical Name:

1. *Capsicum Annum* L.(Fruits are long or small, but pungency is high.)
2. *Capsicum frutescens* L. (Large and inflated variety with very little pungency.)

Family:

Solanaceae.

Chilli is one of the most valuable crops of India. The crop is grown largely for its fruits all over the India. It is used in India as a principle ingredient of various curries, and chutneys. It is also used for vegetables, spices, condiments, sauces and pickles. Dry chillies are used for curry powder. Red colour in chili is due to "Capsanthin". Pungency in chillies is due to the active constituent "Capsaicin", an alkaloid, is extracted from chillies and is used to medicine

Package of Practices of Chilli or Capsicum Cultivation as Follows:

1. Climate and Soil Requirement
2. Sowing Time and Seed Rate
3. Layout and Spacing

4. Manures and Fertilizers

5. Improved Varieties

6. Irrigation

7. Intercultural operations, Pruning and Staking / Training

8. Harvesting and Yield

9. Insect Pest Control

10. Diseases Management

Climate and Soil Requirement for Chilli or Capsicum Cultivation

Climate:

The chili is a plant of tropical and sub -tropical region -It grows well in warm and humid climate and a temperature of 20°C to 25°C. Low moisture in soil during blossom development and fruit formation causes the bud, deblossom and fruit drops. Excessive rainfall is detrimental to the crops, because it brings about defoliation and rotting of the plant. As a rainfed crop, it is grown in areas receiving an annual precipitation of 25-30 inches.

Soil and its Preparation:

Chilli can be grown in all type of soft but the sandy - loam, clay loam and loam soils are best suited for it, the soil must be well drained and well aerated. Acidic soils are not suitable for chili cultivation.

The land is prepared by giving 2-3 ploughings and clod crushing after each ploughing. Compost or FYM @ 150-200 quintals should be spread and mixed well in the soil at least 15-20 days before sowing. At the last ploughing 0. H. C. @ 8-10 kg per acre of Aldrin or Heftaf @ 10-15 kg per acre should be applied to the soil to protect the crop from white ants and other soil pests.

Sowing Time and Seed Rate for Chilli or Capsicum Cultivation

Sowing Time and Seed Rate:

In case of chilli crop, for kharif it sown in May - June and for summer crop, it is sown in the month of January. 1 to 1 1/2 kg seed is required for 1 hectare area.

Layout and Spacing for Chilli or Capsicum Cultivation

Layout & Spacing:

Ridges and furrow type of layout is used. Seedlings are raised on raised bed. Spacing for rainfed crops is 60 x 45 cm & for irrigated crops is 60 x 60 cm.

Manures and Fertilizers for Chilli or Capsicum Cultivation

Manures & Fertilizers:

9 to 10 tones / hectare of FYM or compost is applied at the time of field preparation. In chilli for rainfed crop 50 kg N and 25 kg P should be applied 1/2 dose of N full dose of P applied at the time

of transplanting. Remaining Yi dose of N applied 30 days after transplanting for irrigated crop 100 kg N, 50kg P & 50 kg K should be applied per hectare. Fertilizers are applied in four equal doses. First applied at the time transplanting remaining doses are applied at 4th, 11 & 13' week after transplanting.

Improved Varieties of Chilli or Capsicum

Improved varieties: N.P. 46, JwaJa, G-3, CA-960, Pant C-t X - 235, AKC - 79-18, Parbhani Tejas. Local: Achalpuri, Bhavapuri, Vashirn, Malkapuri.

Chilli: (*Capsicum annuum*)

1. Agnirekha:

It is a derivative of the cross Dondaicha x Iwala Released for green fruits by MPKV, Rahuri in 1992, for summer season. Further it is also recommended for cultivation by Project Directorate of Vegetable Research in 1996 for Zone VII. Plants tall and spreading, fruits long, bold, smooth and light green in colour. Duration is 180 days. Average yield of green" fruit is 250 quintals per hectare. Average yield of dry chilli is 27 quintals per hectare. (MPKV, Rahuri).

2. Musahvadi:

Developed by selection from local collection of village Musahvadi, Dist. Ahmednagar (M.S.). It is released in 1988 by MPKV, Rahuri and also by the Central Sub Committee on Crop Standards, Notification and Varietal Release for Zones IV, V, VII and VIII in 1987 at the national level. Plants are tall, spreading habit with dark green foliage. Fruits are smooth and medium long. Mature fruits dark green with black patches, ripe fruits dark red with retentive colour. Duration is 180 days. It is tolerant to die back and wilt diseases. Average yield of red dry chilli fruits is 19 quintals per hectare. (MPKV, Rahuri).

3. Phule Jyoti:

It is released in 1995 for kharif season. Plants are tall with spreading and branching habit from ground level. Leaves broad and dark green in colour. Fruits are smooth, medium long, borne in cluster of 5-6 with pendent habit. It is tolerant to thrips, mites and wilt. Average yield of green fruits 250 quintals per hectare and that of red dry fruits is 24 quintals per hectare. (MPKV, Rahuri).

4. Pbiile Sai:

Developed by selection from the cross Pani C-I x Kamandalow. It has high yield potential under rain fed conditions. It retains the colour in storage for 6-7 months. It is moderately resistant to thrips and anthracnose (MPKV Rahuri). %. Phule.

5. Suryamukhi:

It is released in 1996. Plants tall, spreading and branching habit from ground level, leaves broad and dark green. Fruits smooth, medium long, borne in cluster of 5-6 and upright erect. It is tolerant to thrips and wilt. Average yield of green fruits is 190 quintals per hectare and yield of red dry fruits is 22 quintals per hectare. (MPKV, Rahuri).

6. Phule Mukta:

It is developed by employing pure line selection and released by MPKV Rahuri. It has dark green foliage, plants tall and medium spreading. Fruits are small, dark green and smooth. Ripe fruits are dark red in colour. Duration is 210 days. Resistant to powdery mildew and Fusarium wilt. It is tolerant to leaf curl, thrips and mites and resistant to lodging, suitable for both the kharif and summer seasons. Average yield of dry chilli is 23 quintals per hectare. MPKV, Rahuri).

7. Surkta:

Surkta has been released for Vidarbha region both for the rain fed conditions of kharif season. It produces bright deep red fruits. Fruits are very hot in taste. It gives about 35 per cent higher fruit yield compared to variety CA-960. Yield of red ripe fruits is 68 quintals per hectare. (PDKV, Akola)

8 Jayanti:

It is notified during the year 1996-97. It has been released for Vidarbha and at the national level under irrigated conditions. Plants are medium to tall in height with light green foliage. Fruits are medium to long in length (9-10 cm) with notch on 1/3rd portion from the tip of the frill. Fruits are pale green, white — green and turn red on ripening. Yield of red ripe chilli is 18-20 quintals per hectare. (PDKV, Akola)

9. Parbhani Tejas:

It is released in 1992 for Marathwada region for rain fed as well as irrigated conditions for green as well as red dry fruit production. It has pungency, fruits are longer (maximum 22 cm). Yield of dry red fruit is 18 quintals per hectare. Average yield on 20 farmers' field has been found 11.56 quintals per hectare as compared to 7.63 in case of NP-46. (MAU, Parbhani)

10. Kankan Kirti:

Plants are dwarf with 50-60 cm height and dark green leaves. Dark green fruits turn red when mature, mild pungent. It yields 12-14 tones per hectare of green chilli. Fruits are dark green, lustrous, good keeping quality and good export potential. (BSKKV, Dapoli)

Irrigation for Chilli or Capsicum Cultivation

Irrigation:

Chilli is grown both as rain fed and irrigated crop. First irrigation is given after the transplanting and subsequent irrigations are given 5-7 days interval depending on weather and conditions of soil during summer and rainy season and after .Every 10 to 15 days in winter. The maintenance of uniform soil moisture is essential to prevent blossom .and fruit drops. Bosaocn and Garibaldi (1971) reported that the irrigation when soil moisture tension exceeded one atmosphere increased the yield of the unshared plant, but the number of fruits was unaffected. Flower drop in chillies is a great problem and it depends on the high temperature, low moisture availability, shading and light intensity. Flowers and /or flower bud abscission was increased under short day (day length 12 hours) and high temperature (28 °C to. 33 °Q. Spraying the chilli with Planofix (NAA) at 10 ppm at flower initiation stage and 15 days later (peak flowering) reduces the flower drop and increases the fruit set

Intercultural Operations in Chilli or Capsicum Cultivation

Interculture Operations:

2-3 shallow hoeing should be given to the soil to kill the weeds and provide soil mulch during early stages of growth- Application of weedicides for controlling the weeds is found effective. Lasso @ 1.5 litre per hectare with one hand weeding or Tok-E 25@ 2 litres per hectare with one hand weeding were effective in controlling the weeds.

Harvesting and Yield in Chilli or Capsicum Cultivation

Harvesting:

1. Chillies which are used for vegetable purposes are generally harvested while they are still green but full grown.
2. Chillies are harvested at red stage for caning purpose. Chillies used for drying are harvested at full ripe-stage.

Yield:

The yield varies according to the system of cultivation. The yield of dry chillies of rain fed crop is 200 - 400 kg and that of irrigated crop is 600 - 1000 kg per acre. The proportion of dry to fresh japed chillies varies from 25- to 40 per cent.

Insect Pest Control in Chilli or Capsicum Cultivation

Insect Pests of Chilli:

1) Chilli Thrips:

Thrips is the common pest which effects the crop throughout its life cycle. But they are more severe when plants begin to flower. These small insects suck the sap from the foliage and lacerate the leaf tissue, which result in curling of leaves and fall down of flowers prematurely.

Control Measures:

Thrips can effectively be controlled by spraying carbaryl 50 W @ 3 gm or Zolene @ 3 ml or Dimethoate (Rogor 30 EC) or Monocrotophos (Monocil) tgi 1 ml per litre of water at fortnightly interval

2) Pod Borer:

The caterpillar eat leaves and later on bores the pod, which result in the deterioration of quality and market price of the product.

Control Measures:

The control measures are timely spraying the crop with quinalphos (Exalux-25 EC) @ 4 ml or carbaryl 50 W, (Sevin 50 W) @ 3 gm per litre of water, starting from flower bud formation.

3) Aphids:

Aphid suck the sap from the plants; they generally attack the crop in winter months and at the later stages of the crop. The quality of the produce is spoiled by imparting blackish colour to the Calyx and pods. They also serve as a Vector to Virus,

Control Measures:

The aphids can effectively be controlled by spraying the crop with Dimethoate (Rogor-30 EC) or Methyl Parathion (Metacid 50 EC) @ 1m (or Phosphamidon (Demicron-100 EC) @ 0.5 ml per litre of whiter.

Diseases Management in Chilli or Capsicum Cultivation

Diseases of Chilli:

1) Damping Off:

It is a serious disease of brinjal seedlings and mainly occurs in nursery bed. The disease infected seedlings rot at ground level and then the plants fall over ground. The seedlings die in patches.

Control Measures:

1. The seed bed should be treated with Formalin before sowing of seeds.
2. The seeds should be treated without water (30 minutes at 520 C) or Cerasan or Agrosan G.N. before sowing of seed.
3. The seedlings in the nursery should be sprayed with any fungicides at a regular interval.

2) Bacterial Leaf Spot:

Small dark, greasy spots are formed on leaf, petiole and tender parts, of the plant. Water soaked spots appear on green fruits. In severe cases the leaf may drop off and cause considerable loss to the crop.

Control Measures:

Spraying Agrimycin - 100 k at 200 ppm plus copper oxychloride 0.3 per cent controls the disease effectively.

3) Anthracnose:

Dark sunken spots are formed on fruits and pink or dark coloured dots appear in the centre of the sunken spots. Due to this spots, the fruits rot and fall. The fungus may cause "Die back" of the twigs also. Die back disease attacks mainly the upper portion of the plants spreading gradually from the top to downward; as a result the branches dry up. Moist weather, shade and heavy dew favour the occurrence of the disease.

Control Measures:

The control measures are the treatment of seed with Cerasan before sowing, removing and burning of attacked plants or branches and spraying the disease affected crops with Mancozeb (Dithane M-45) @ 2.5 gm per litre of water.

4) Leaf Curl:

The disease affected leaves becomes small in size accompanied by downward curling. The leaves may fall off in case of severe attack. The disease usually spreads through insect vectors such as thrips and aphids etc.

Control Measures:

Control of insect vectors by spraying the crop with Dimethoate (Rogor -30 EC) or Monocrotophos (Monocil) @ 1 ml per litre of water indirectly helps to check the spread of this disease.

Cultivation of Pumpkin (Cucurbit moschoto poir)

Botanical Name:

Cucurbit moschoto poir.

Local Name:

Kashiphal –Hindi, Lal shopla- Marathi.

Family:

Cucurbitaceus.

It is a favorite and popular vegetable of many people during summer and rainy season. Pumpkin is one of the most important cucurbitaceous vegetable crops. The ripe pumpkins are used as cooked. It is one of the main vegetable in any festival. It is round or long in shape.

Climate:

1. It is warm season crop. But it can withstand cool weather. The plant is also tolerant to partial shade.

Soil and its Preparation:

1. Pumpkin is grown on all types of soil. But loam, sandy loam and clay loam soils are considered best for its cultivation.

2. The soil should be thoroughly prepared. It does best at a Ph of 6.0 or 6.5.

Sowing of Seed:

Two crops are raised commonly in the plains during January – March and June – July. Seeds are sown in well prepared pits at a spacing 3 X 3 m. about 4-6 kg. of sees are required to raise one ha. Layout ring and basin is used.

Manures and Fertilizer:

30 to 50 cartload well decomposed FYM should be added at the time of soil preparation. 50 kg N, 25 kg P/ha. Should be given ½ N and full phosphorus should be given at the time of sowing seeds and remaining half N should be given 30 days after sowing.

Intercultural Operation:

One or two weeding may be done during early stage of growth. No irrigation is given to rainy season crop. The summer season crops should be irrigated after third or fourth day.

Harvesting and Yield:

Pumpkin is harvested either in green or mature stage according to market demand. The fruits can be stored for a few months provided they are kept in well ventilated rooms. The yield may range from 15 to 25 t per ha.

Varieties:

Co.1, Co.2, Arka Suryamukhi, Arka chanada, HR 83-1-1.

Cultivation of Bottle Gourd or Lauki or Calabash cucumber or White Flowered Gourd (Lagenaria steraia)**Botanical Name:**

Lagenaria steraia.

Local Name:

Dudhya Sweet- Marathi, Lauki- Hindi.

Family:

Cucurbitaceae.

Bottle gourd is a very important crop in India. The fruit in green stage and leaves with stem are used as vegetable. The hard shell of the fruit is used for different purpose.

Climate:

Hot and moist climate is favorable for its cultivation. It cannot withstand frost.

Soil and its Preparation:

Bottle gourd can be grown in any types of soil. But sandy loam soils are best suited for its cultivation. The land should be prepared thoroughly by five to six ploughings.

Time of Sowing and Layout:

The seed is sown from January to end of February for summer crops. June – July for rainy season crop in the plains and April in the hills. Layout is ring and basin used.

Seed Rate:

The seed rate is 3 to 6 kg/ ha.

Methods of Sowing and Spacing:

The seed is sown by dibbling method at spacing of 2 to 3 X 1.0 to 1.5 m. Generally three to four seeds are sown in a pit at 2.5 to 3.0 cm depth. Manures and fertilizer same as for pumpkin.

Intercultural Operation:

Two to three hoeing is given to keep down the weeds during the early stage of growth. The rainy season crop is usually stalked, often trained on a bower made of bamboos and sticks.

Irrigation:

The summer crop requires frequent irrigation at an interval of 4 to 5 days. The winter crop is irrigated as and when needed.

Harvesting:

The fruits should be harvested when they are still green. Delay in harvesting causes the fruit to become unfit for marketing.

Yield:

The average yield is 90 to 120 quintal / ha.

Varieties:

There are some varieties of bottle gourd, viz summer prolific Round, summer prolific long, Pusa Navven, Arka Bahar, Pusa Manjiri, Pusa Meghdoot etc.

Bottle Gourd: (*Lagenaria sicerat*)**1. Samrat:**

Development by selection from local germplasm collected village Dahanu, Dist. Thane of Maharashtra and released in 1992. Fruits are 30 -40 cm long, green in colour with dense pubescence and cylindrical in shape. Good for box, packing. Good keeping quality. Duration is 180-200 days. Average yield is 430 quintal per hectare.

Cultivation of Bitter Gourd or Karela (*Momordica charantia*)**Botanical Name:**

Momordica charantia L.

Local Name:

Karela - Hindi; Karli - Gujarati & Marathi, Paval

Family:

Cucurbitaceae.

Bitter gourd is one of the most popular cucurbitaceous vegetable. It is grown extensively throughout India; the fruit is wormicidal and good for rheumatism.

Climate:

It is a warm, season crop. Hot and moist weather is favorable for its growth and development. Low temperature inhibits the germination of seeds. It grows best at temperatures between 18°C and 24°C.

Soil and its Preparation:

Bitter gourd can be grown in all types of soil. But loam and silty loam soil is more suitable for its cultivation. It grows well on silty soil on river beds. The land should be well prepared.

Sowing of Seeds:**Time of Sowing:**

The seed is sown from January to March for summer season crop, June-July for rainy season crop in the plains and March to June in the hills.

Seed Rate:

The seed rate is 4 to 5 kg/ha.

Method of Sowing:

The seed is sown by dibbling method at a spacing of 120x90 cm. Generally three to four seeds are sown in a pit at 2.5 to 3.0 cm depth. The seeds are soaked in water over night before sowing for better germination. Seed germination was enhanced by soaking the seeds for 24 hours in solution of 25 to 50 ppm GA and 25 ppm boron. In flatbed layout seeds are dibbled at the spacing 1 x 1 m.

Manuring:

30 to 50 cartload well decomposed FYM should be added at the time of soil preparation 50 kg N, 25 kg P/ha. Should be given ½ N and full phosphorus should be given at the time of sowing seeds and remaining half N should be given 30 days after sowing.

Intercultural Operation:

Two to three hoeing is given to keep down the weeds during the early stage of growth. The rainy season crop is usually staked, often trained on a bower made of bamboos and sticks.

Harvesting:

Harvesting is done when the fruits are still young and tender at every alternate day. Picking should be done carefully so that the vine may not be damaged. The fruits should not be allowed to mature on the vines. The harvested fruits may be stored for 3 to 4 days in-cool condition.

Yield:

The yield is 60 to 100 quintal/ha.

Varieties:

There are some varieties of bitter melon viz. Pusa Domousmi, Coimbatore, Long, Arka Harit, Long green etc.

Bitter Melon (Momordica charantia)

1. Hirkani:

Developed by selection from local germplasm and released in 1991 for western Maharashtra Fruits is dark green, 15-20 cm with prickles. Duration is 160 days. Average yield is 138 quintals per hectare. (MPKV, Rahuri)

2. Phule Green Gold:

Developed by selection from cross Green Long x Delhi Local and released in 1996. Fruits dark green, 25-30 cm long and prickled. Crop duration is 150-180 days. Average yield is 230 quintals per hectare. It is tolerant to downy mildew. Besides it is released through State Variety Release Committee of Maharashtra. It is also notified by the Central Sub-Committee on Crop Standards, Notification and Varietal Release at the National level. (MPKV, Rahuri)

3. Phule Priyanka:

It is a F1 hybrid released by MPKV, Rahuri, suitable for both the kharif and summer seasons. Fruits dark green, fruit surface is highly prickled. Fruit length is about 20 cm. Average yield is 200 quintals per hectare. It is tolerant to downy mildew. Responds well to bower system of training. Besides it is released through State Variety Release Committee of Maharashtra. It is also notified

by the Central Sub Committee on Crop Standards. Notification and Varietal Release at National level. (MPKV, Rahuri)

4. Konkan Tara:

It gives fruit yield about 15-20 tones per hectare. Fruit possess dark green prickles, medium long and spindle shaped having good keeping quality and suitable for export purpose. It is recommended for the Konkan region (BSKKV, Dapoli).

5. Phule Ujwala:

Foliage is dark green and stem is green in colour. Average fruit weight is 84 g. Fruit length is 18 cm and diameter is. 4.5 cm. Duration is 180 days. Fruits are highly prickled. Average fruit yield is 174 quintals per hectare. (MPKV, Rahuri)

Cultivation of Ridge Gourd or Ribbed Gourd (*Lufa acutangula*)

Botanical Name:

Lufa acutangula roxb.

Local name:

Jhina turai- Hindi, Sirola- Gujarati and Marathi.

Family:

Cucurbitaceae.

Ridge gourd can be grown through the year and are used as vegetables. It is more popular vegetable in the south and east India.

Soil and its Preparation:

Ridge gourd can be grown in all kinds of soil, loam, clay loam and silt soils are best suited for its cultivation. The land is prepared by ploughing for three to four times, followed by planking.

Sowing of Seed:

a) Sowing Time:

There are two types of ridge gourd commonly grown in our country and the seed is sown in different times according to type of crop.

b) Seed Rate:

The seed rate is 4 to 5 kg/ha.

c) Method of Sowing:

The seed is sown by dibbling method at a spacing of 1.5 to 2.0 X 1.0 to 1.5 m. Two to three seeds are sown in each pit. Layout is ring and basin.

Manuring:

30 to 50 cartload well decomposed FYM should be added at the time of soil preparation. 50 kg N, 25 kg P/ha. Should be given ½ N and full phosphorus should be given at the time of sowing seeds and remaining half N should be given 30 days after sowing.

Intercultural operation:

Shallow cultivation should be given during the early stage of growth. The plants should be provided a suitable support made of bamboo sticks.

Irrigation:

The summer crop should be irrigated just after sowing and subsequent irrigation is given at four to five days interval. No irrigation is given in rainy season crop.

Harvesting:

The fruit become ready for harvest from 55 – 60 days of sowing. The full grown tender fruit should be harvested at weekly interval by cutting them with a knife.

Yield:

The yield is about 75 to 100 quintal/ ha.

Varieties:

There are some varieties of ridge gourd recommended for cultivation in our country. Pusa Nasdar, Desi Chaitali, Co- 1.

Ridge gourd: (*Luffa acutangula*).

1. Kankan Harita:

It yield about 150- 200 quintal per hectare. Fruits are long (40 cm) uniform soft ridged and firm flesh. It is an early variety suitable for both the Kharif and summer seasons. A fruit weights about 200 g. (BSKKV, Dapoli).

2. Phule Sucheta:

It is developed by employing a pure line selection. Fruits are green, slender, tender, and medium in length (33.5 cm) with prominent ridges. It shows filed tolerance to downy mildew. Suitable for in both the Kharif and summer season of western Maharashtra. Average fruit weight is 118 g. A vine produces 18 fruits. Average yield is 119 quintal per hectare.

Cultivation of Cucumber or Kakdi (*Cucumis sativus*)

Botanical Name:

Cucumis sativus L

Local Name:

Kakdi-Maraihi; Kakkari-Malayalam; Kamal Kakdi-Hindi.

Family:

Cucurbitaceae.

Cucumber is an important summer vegetable commonly grown throughout India. Cucumber plant, as creeper has a climbing or trailing habit. The tender fruits are eaten raw or with salt in salad. They are also used as cooked vegetable. It is reported that the oil extracted from seeds is good for brain and body. Drinking of water immediately after eating cucumber should be avoided, as it sometimes causes severe indigestion.

Climate:

The cucumber is a warm season crop and grows best at a temperature between 18°C and 24°C. It does not withstand even light frost. Soil and its preparation :Cucumber can be grown in all types of soil from sandy to heavy soils. Loam, silt loam and clay loam soils are considered best for getting higher yield. Soil pH between 5.5 and 6.7 is favorable for its cultivation. The land should be prepared thoroughly sowing of seeds.

a) Sowing Time:

The cucumber is cultivated both as a summer and rainy season crop and the seed is sown according to type of crop.

Summer Crop : January to February.

Rainy crop: June-July.

The seed of cucumber is sown in April in the hills.

b) Seed Rate:

The seed rate is 2.5 to 4 kg/ha.

c) Method of Sowing:

The seed is sown by dibbling method at a distance of 1.5 to 2.5 meters (row to row) x 60 to 90 cm. (Plant to plant). Two to three seeds are sown in each pit. Layout in Ring & basing method.

Manuring:

30 to 50 cartload well decomposed FYM should be added at the time of soil preparation. 50 kg N, 25 kg P/ha. Should be given $\frac{1}{2}$ N and full phosphorus should be given at the time of sowing seeds and remaining half N should be given 30 days after sowing.

Intercultural Operations:

Shallow cultivation should be given during the early stages of growth to control the weeds. Herbicides (such as Glyphosphate, Paraquat and Chloramben @ gm per acre) may be used to control weeds.

Irrigation:

The summer crop cultivated as an irrigated crop requires enough soil moisture during its growth and development. So irrigation should be given as and when needed. No irrigation is given in rainy season crop.

Harvesting:

The rail grown tender fruits should be harvested at an interval of two to four days. Timely picking is more important in regard to quality. Yield: The average yield is about 60 to 15 quintals/ha,

Varieties:

Japanes Long green, Balam Khira Peons, priya, Pusa Snajog, Cucumber (*Cucumis sativus*).

1. Himangi:

Developed by selection from the cross Poinsett x Kalyanpur Ageti and released in 1992. Fruits are white in colour, 12-15 cm long and resistant to bronzing. Total duration is 100-110 days. Average yield is 158 quintals per hectare.(MPKV, Rahuri)

2. Phule Shubhangi:

It is released by MPKV, Rahuri. Green skin colour. It is tolerant to powdery mildew. It yields more than vars. Poona Khira and Himangi. Besides it is released through State Variety Release Committee of Maharashtra. It is also notified by the Central Sub Committee on Crop Standards, Notification and Varietal Release at the National level.(MPKV, Rahuri)

3. Phule Prachi:

It is agynoeious F hybrid developed by exploiting a parthenocarpic tropical gynoeious female parent. It is suited to the tropical conditions. Performs well under both the open field and green house conditions. Responds well to the improved agro-techniques like high fertilization, drip irrigation, fertigation, staking etc. Fruit skin is yellowish white in colour suitable to the consumers' preference of Maharashtra. Gives very high yield (average of 18 trials is 359 q/ha i.e. 85% higher than check). Fruits are straight and longer. Fruits are set parthenocarpically. It gives more than 1 000 quintals per hectare yield under staking cum spacing (30 cm x 30 cm) conditions and about 750 quintals per hectare under naturally ventilated green house. (MPKV, Rahuri)

4. Sheetal:

It gives about 250 - 300 quintals per hectare fruits. They are green, cylindrical and long with good keeping quality. Suitable for both the kharif and summer seasons. It also does well in high rainfall areas. It is released for commercial cultivation in Konkan region. Average weight is 200-250 g per fruit. It gives an average yield of 300-350 quintals per hectare. (BSKKV, Dapoli)

5. Phule Champa:

K is a gynoeious fi hybrid suited to the tropical conditions. It is recommended for the open field as well as green house conditions. It has high yield potential. Average yield is 378 quintals per hectare. Fruits are light green in colour, straight and longer. Suitable for all over the country. (MPKV, Rahuri).

Cultivation of Watermelon or Kalingad (*Citntlltts vulgaris* Schard)

Watermelon

Botanical Name:

Citntlltts vulgaris Schard.

Local Name:

Tarbnj and Jamaika- Hindi; Kalingad - Marathi; Taannir mathan or Mandeki patak - Malayalam.

Family:

Cucurbitaceae.

The watermelon is a common summer crop: It is generally grown for its ripe and well mature fruit the fruits are delicious and sweet when ripe.

Climate:

Watermelon needs a long growing season with a relatively high temperature. It can not withstand frost. The seed does not germinate satisfactorily below 21 °C. Dry weather during ripening with high temperature, enough sunshine and dry winds favour the development of flavour and high sugar content.

Soil and its Preparation:

Deep sandy loam soil is best suited for water melon crop. It is generally cultivated on river beds. The optimum soil pH is 5.5 to 7.0. The land is thoroughly prepared by giving 5 to 6 ploughings.

Sowing of Seeds:

a) Sowing Time:

The seed is generally sown from November to March in the plains and from middle of February to end of March in the hills. The seed is soaked overnight before sowing.

b) Seed Rate:

The seed rate is 3 to 4 kg/ha.

c) Method of Planting:

1) Shallow Pit Method:

In this method, pits of 60 x 60 x 45cm. are dug at a distance of 1.5 to 2.5 meter. Each pit is filled with a mixture of soil and 4 to 5 kg of FYM or compost, 30 to 40 gm. Urea, 40 to 50 gm. Single Super phosphate and 80 to 100 gm of muriate of Potash. After filling the pit circular basin are made and five to six seeds are sown per basin at 2 to 3 cm. deep and covered with fine soil, F.Y.M. or compost.

2) Deep Pit Method:

This method is followed in case of river bed cultivation. Circular pits of 60 to 75 cm. diameter and 1.0 to 1.5 m. depth are dug at a distance of 1.5 to 2.5 meter distance both ways.

3) Ridge Method:

The seeds are sown on the side of raised beds.

4) Mound Beds:

Seeds are sown on the mounds. The seed rate is 1.6 kg per acre. The normal spacing is row to row 300 cm and plant to plant 90 cm. The germination percent and the storage life of seeds can be improved by treating freshly extracted seeds with 2 percent HCl or NaOH.

Manuring:

Apply 30-50. Cart loads FYM at the time of soil preparation 80 kg N, 40 kg P & K is recommended by PKV Akola. 1/2 dose N, full dose of P and K should be applied at the time sowing. Remaining half Nitrogen applied one month after transplanting.

Intel-cultural Operation:

Two or three hoeing may be done during the early stage of growth to keep down the weeds and to conserve soil moisture. Two or three plants are kept at one place in the hills or beds by thinning the extra plant. Herbicides (e.g. Chloroxuron and Prometryne @ 3 & *2 kg. a.i. per ha, respectively may be useful for controlling weeds.

Irrigation:

The crop is watered by pitcher or water can after sowing. The Crop is watered sparingly once every seven to ten days.

Harvesting:

The plant bears fruits in 4 to 5 months after sowing. The watermelon should be harvested at the proper stage of maturity. The following points give an indication or ripeness:

1) Withering of Tendrils:

The tendril that subtends the peduncle of the fruit, usually wither when the fruits ripen and watermelon is seldom ripe if the tendril is still green. But it may be unripe even if the tendril is dried in some varieties.

2) Thumping:

Ripe fruit, when thumped with the finger, gives out a muffled dull or dead sound, whereas the immature fruits give out a metallic and ringing sound.

3) Colour of the Ground Spot:

It is white when fruits are green and they rest over it. When the fruits are ripe they change to yellow.

4) Preserve on the Fruit:

A ripe melon emits a crisp cracking noise on being pressed with the flat of the hand. Moreover, the rind of a ripe melon yields little under the pressure of the thumb. In some varieties even sunlight pressure will crack the fruit.

If the fruits are allowed to over ripen, the pulp loses the sweetness and soon develops an off flavour and rot. It is, therefore, very necessary that they should be harvested at the full ripe stage, because they ripen very little on their way to the market.

Yield:

The average yield is 200 to 300 quintal/ha.

Varieties:

There are a large number of varieties grown all over the country Following are the important varieties of watermelon. Asahi Yamato, Sugar baby, Arka Manik, Ark'a Jyoti, Durgapur Meths, Nem Hamshire Midget, Pusa Rasaal, Durgapura Kesar, Early Kansas.

ultivation of Muskmelon or Kharbooj (Cucumis Mela)

Botanical Name:

Cucumis Mela .L.

Local Name:

Kharbooj- Hindi, Kharbuj- Marathi.

Family:

Cucurbitaceae.

Muskmelon is an important crop cultivated through India. It is used as desert fruit and also eaten alone. Very rarely it is used cooked vegetable in its green stage.

Climate:

The muskmelon thrives best and develops the highest flavor in a hot dry climate. High temperature and sunshine produce melon high in sugar contents. Though sweetness and fine flavor are hereditary characteristics, yet they are greatly influenced by environmental condition. It is highly susceptible to frost; it is grown as summer crop.

Soil and its Preparation:

Muskmelon is mainly grown on sand and sandy – loam soils on river bed. But other type of soil is also favorable for its cultivation. The optimum soil PH is 6.0 to 6.7. The land is prepared thoroughly by repeated ploughing.

Sowing of Seeds:

a) Time of Sowing:

The seeds may be sown from November to March in the plains and April- May in the hills. The seeds are sown in November- December in west Bengal .Bihar and south India. Transplanting of about a week old seedlings may also be done.

b) Seed Rate:

The seed rate is 2 to 3 kg/ha.

c) Method of Sowing:

1. Shallow Pit Method:

In this method pits of 60 X 60 X 45 cm are dug at a distance of 1.5 to 2.5 m. each pit is filled with a mixture of soil and 4 to 5 kg of FYM in compost, 30 to 40 gm. Urea, 40 to 50 gm. Single phosphate and 80 to 100 gm of muriate of potash. After filling the pit circular basin are made and five to six seeds are sown per basin at 2 to 3 cm. deep and covered with fine soil, FYM or compost.

2. Deep Pit Method:

This method is followed in case of river bed cultivation. Circular pits of 60 to 75 cm. diameter and 1.0 to 1.5 m. depth are dug at a distance of 1.5 to 2.5 m distance both ways.

3. Ridge Method:

The seeds are sown on the side of raised beds.

4. Mound Method:

The seeds are sown on the mounds. The seed rate is 1.6 kg per acre. The normal spacing is row to row 300 cm and plant to plant 90cm. the germination percent and the storages life of seed can be improved by treating freshly extracted seeds with 2 % HCL or NaOH.

Manuring:

Apply 30 -50 cart loads FYM at the time of soil preparation 80 kg N, 40 kg P and K is recommended by PKV Akola, 1/2 dose N, full dose of P and K should be applied at the time sowing. Remaining half nitrogen applied one month after transplanting.

Intercultural Operation:

Two or three hoeing may be done during the early stage of growth to keep down the weeds and to conserve soil moisture. Two or three plants are kept at on place in the hills or beds by thinning the extra plant.

Irrigation:

Irrigation is given as and when it is necessary. Crops sown on river beds or not needs and irrigation.

Harvesting:

Muskmelon fruit will be ready for picking in about 110 days depending upon variety and agro-climate. The stage of maturity is generally judged by the change in the external colour of fruit, softening of the rind and development of abscission layer. In muskmelon when fruit are mature slips out easily from vine. The stage at which the fruit are picked, also affects the quality of fruits, hence they should be picked in time.

Yield:

The average yield is 100 to 150 quintal / ha.

Varieties:

Followings are the important varieties of muskmelon. Arka rajhans, Hara Madhu, Pusa Asaarbati, Arka Jest, Durgapura Madhu, Punjab sunahari, Lanow safed, Annamalai, Haribhari.

Insect Pest Management of Cucurbitaceous Crops

1. Red Pumpkin Beetle:

It is the most serious insect pest of cucurbits. The larvae and adult of this pest cause damage by eating away the young leaves and flowers at the seedling and flowering stage respectively. The grub bores into the roots and fruits lie on the ground and pupate in the soil.

Control Measures:

- i) Hand pickling and dusting keratinized ash is the most common method of controlling this pest.
- ii) Spraying the crop with Malathion 50 EC or Dichlorovus (Nuvan, Vapona etc) @ 2 ml per liter of water can control this pest.

Fruit Fly:

The maggots of this pest enter the fruit and feed on the pulp, as a result of which it starts rotting.

Control Measures:

- i) Collection and destruction of all infested fruit in the field will help in minimizing the damage to the crop.
- ii) Dusting the crop with Malathion or Dipterex Powder in the affected field will help in controlling adult fly.
- iii) Use of bait traps prepared from protein hydrolyzate @ 450 gm, Malathion or Dipterex water dispersible powder (25 %) @ 450 gm and water @ 1 gallon is most effective of controlling of this pest.

Diseases Management of Cucurbitaceous Crops

1. Powdery Mildew:

Tiny white superficial spots appear on leaves and stem and they become powder on enlarging. Due to attack of this disease, premature defoliation occurs and plants remain stunted in growth. The fruit too do not set or remain smaller in size.

Control Measures:

- i) This disease can be controlled by dusting sulphur (sulfex) or spraying Karathane or Moresta @ 2 ml per liter of water.
- ii) Abol- waja et al. reported that systematic fungicide nimro at 0.05 per cent, Saprol at 0.1 % Calixin at 0.04 % and Ben late 0.05 % gave better control of powdery mildew in cucurbits.

2. Downy Mildew:

Purplish spots appear on lower surface and yellow spots on upper surface of leaves; fruit do not mature or do not contain right flavor.

Control measures: application of fungicidal spray such as Dithane Z-78, Dithane M- 45, Blitox etc) once a week helps to control this.

3. Anthracnose:

Watermelon muskmelon and cucumber is mainly affected by this disease. Light brown spots are formed on leaves which turn to dark brown and then to red, and the leaves present a scored

appearance. Circular to oval sunken lesions develop on the disease infected fruits and the fruit rot later on.

Control Measures:

Crop rotation with non – cucurbit crops, use of disease free seeds, seed treatment and spraying of fungicides are the recommended control measures of this disease.

4. Fusarium Root Rot:

Sudden mid season wilt base of stem becomes dark brown, with a soft mushy cuticle wilt, fruit rot, and young seedling damp off.

Control Measures:

Crop rotation and seed treatment with hot water at 55°C for 15 minutes followed by treatment with chloride (1 in 1000) are the recommended control measures of these diseases.

5. Mosaic:

Mottled leaf with roughened surface occurs. The green colour completely lacking in fruit. In severe cases the plants are yellow and dwarfed and bear little or no fruit. Aphids transmit this disease.

Control Measures:

Growing resistant variety and spraying the crop with Dimethoate (Rogor-30 EC) or Methyl parathion (Metacid – 50 EC) or Oxidomethion methyl (Metsyton – 25 EC) 2 ml per liter of water for controlling the insect vector are the recommended control measures of this disease.

Cultivation of Pea (*Pisum sativum*)

Botanical Name:

Pisum sativum L.

Family:

Leguminaceae

Pea is an important vegetable in India; the crop is generally cultivated for its green pods. It is highly nutritive and is rich in protein. It is used as a vegetable or in soup, canned frozen or dehydrate. It is cooked as a vegetable along or with potatoes. Split grains of pea are widely used for dal. Pea straw is a nutritious fodder.

Climate:

Pea is a cool season crop and performs best at 10 °C to 18 °C. The flower and young pods are badly affected by frost. The germination of seeds takes place at 3.3 °C soil temperature. Boswell (1920) reported that as the temperature increases during the growing season the yield decline sharply. The optimum mean monthly temperature for pea is 12.8 °C to 18 °C.

Sowing of Time:

a) Sowing Time:

The pea is generally sown in India in Rabi season from the beginning of October to mid of November in the plains and from middle of March to end of May in the hills. Sowing of seed during the first week of November is proper time to get higher yield.

b) Seed Rate:

The optimum seed rate is 25 to 30 kg/ha.

C) Method of Sowing:

The pea is generally sown by broad casting. But it may also be sown by dibbling or behind the plough. The seeds are soaked in water overnight before sowing for better germination. Seeds treated with Rhizium culture give higher yield. Pal Sodkar et al. (1974) reported with soaking of seeds in G. A 10 ppm for 12 hour gave the highest germination and yield.

Layout and Spacing:

Flat bed layout is used. Spacing is 45 X 20 cm.

Manuring:

30 to 50 cartload should be applied at the time of soil preparation. 25 kg N, 50 kg P and 50 kg k/ha should be applied ½ dose or full dose P and K should be applied at the sowing. Remaining half nitrogen applied one month after sowing.

Intercultural Operation:

a) Trailing and Staking:

This is an important operation to be done when vines were about two month old and are at spreading stage. Generally the plants should be supported on bamboo sticks. Delay in this operation will reduce the yield considerably.

b) Weed Control:

it is very difficult to control the weeds of pea field by mechanical methods as the crops are sown in rows in closed spacing. Uses of herbicides have been proved very such effective. Atrazine, propazine and simazine at 0.54 kg per acre gave good broad leaf weed control and late control of wild oats damage the peas. Prometryne @ 400 gm per acre was most effective in improving vegetative growth and yield of pods.

Irrigation:

Water requirement of pulse crops are higher than cereals. The water requirement of pea depends largely on Agronomic condition of the locality. The crop may be irrigated at both 10 days interval. Where rainfall is low, peas irrigation is very necessary at flowering and grain development stage.

Harvesting:

Peas are harvested for table use when the pods are will fill and the young tender peas changing in colour from dark to light green. Peas may be picked in 45 to 60 days, 75 days and 100 days according to early. Mid season and late. Airtimes respectively, 3 to 4 pickling are done within the interval of 2 to 10 days. Fresh unshielded peas may be kept two at 0 C and 90- 95 percent relative humidity.

Yield:

The yield of per hectare varies according to the variety:

Early Variety:

25 to 40 Quintals.

Mid Season and Late:

50 to 60 Quintals.

Varieties:

Varieties according to their period of maturity.

Early Variety:

Early Bejar, Arket.

Mid season Variety:

Bonevilla, Jawahar.

Cultivation of Indian Bean (Dolichos lablab)

Botanical Name:

Dolichos lablab (Roxb) and L.

Family:

Leguminaceae.

Indian bean is primarily grows for its green pod. Young pods are cooked as vegetable. Schaffausen (1963) reported then Indian bean is used for pod, feed and soil improvement.

Climate:

Indian been is relatively a cool season and wrought resistant crop. The plants starts fruiting from the beginning old cold season 2 ½ - 3 months after sowing and continue to give green pods through winter and spring month.

Soil and its Preparation:

Indian peas can grow in a wide range of soils. But loam, silty loam and clay loam soils are best suited for Indian bean. The land is prepared thoroughly by giving 5 – 6 ploughing.

Sowing of Seed:

a) Time of Sowing:

Filed varieties are sown with the advent of the monsoon in July- August, come into flower in Novemeber – December and are harvested in January – February.

b) Seed Rate:

The seed rate is 8 to 10 kg/ha.

c) Method of Sowing:

The seeds may be sown in rows, or on ridges by drilling or by dibbling. Flat bed and ridges and furrow layout is used. Spacing is 90 X 90 cm or 1m X 1m.

Interculture:

2- 8 weeding are required for control of weeds. Spraying insecticides and fungicides from the time to time for control of insect pests and diseases.

Irrigation:

Indian bean is commonly cultivated as rained crop. Irrigation will be given as and when required.

Harvesting:

The Indian bean becomes ready for harvesting after two and half to three months of sowing. Full grown bean is harvested according to the need.

Yield:

The average yield is 100 to 120 quintal of green pods per hectare.

Varieties:

Dasara, dipali, Kankan Bushan, local.

Dolichos Bean (Dolichos Purpureus):**1. Kankan Bushan:**

Green pod yield is 80- 100 quintal per heater. Pods are slightly curved tender and stingless. Bushy growth habit. Inflorescence is white.

2. Dasara:

Notified during the year 1985- 86. Suitable for Kharif season. Pole type growth habit, fruit length 10- 11 cm and breath 2.0 to 2.5 m. fruits are violet in colour. Seeds are milky white, oval shaped with black spot. Average yield is 150- 200 quintals per heater.

3. Deepali:

Notified during the year 1985- 86, suitable for Kharif season. Pole type growth habit, fruits are milky in colour, fruit length 20-25 cm and breath 3.0 to 3.5 cm. seeds are red in colour. Fruits are ready for pickling in 150 days. Yield is 150 – 200 quintal per heater.

4. Phule Gauri:

It is developed by pure line selection. Pods are whitish green in colour, attractive tender flat and slightly curved. It is to the vars. Dasara and Deepali for riled. Average yield is 259 quintals per ha.

It is good for both the Kharif and Rabi season. It is tolerant to leaf miner and pod borer.

Cultivation of French Bean or Kidney Bean (*Phaseolus vulgaris*)

Botanical Name:

Phaseolus vulgaris L.

Family:

Leguminaceae.

French bean is valued as vegetable and seed. It is extensively grown as green vegetable.

Climate:

Kidney bean is day neutral crop. It is very sensitive to high temperature and frost. Soil temperature approximately 32.2 °C (90 °F) is essential for maximum vegetative growth.

Soil and its Preparation:

French bean can be grown or practically in almost all type of soil from light sandy soil to heavy clay soil. The optimum soil pH for getting good crops is 5.3 to 6.0. The land is prepared thoroughly by 5-6 ploughing.

Sowing of Seed:

a) Time of Sowing:

The seed is sown in June – July and January- February in the plains and from March to the beginning of May in the hills.

b) Seed Rate:

The seed rate is 40 to 50 kg/ha.

c) Layout and Spacing:

The seed may be sown by dibbling, drilling and broadcast at a depth of 2.0 to 3.0 cm; the seeds should be inoculated with rhizobial culture before sowing. The flat bed layout is used. Spacing is 45 X 20 cm. recommended by PKV, Akola.

Interculture Operation:

Shallow cultivation is given to keep the crop free from weeds. Herbicides such as dinitro-methyl 2 – 3 kg per acre and sodium salt of pentachlorophenol @ 6 kg per acre as pre-emergence treatment have been effective in controlling the weeds.

Irrigation:

The soil must have sufficient moisture. Irrigation should be given just prior to blooming, during flowering and pod development stage.

Harvesting:

The green pods become ready for harvesting after 45 to 75 days of sowing according to the variety and they should be picked as and when they are ready. The dry beans harvested when a large percentage of pods are fully ripe and most of the remainder has turned yellow. Harvesting should be started before the lower pods dry enough to shatter.

Yield:

Yield obtained 40 – 50 quintal green pods/ ha.

Varieties:

There are numerous varieties in French bean and they can be broadly classified according to the habit into bush and pole beans.

a) Bush bean variety:

Pencil wonder, contender, Pusa parvati, Arka komal, jampa, bountiful, prider.

b) Pole bean variety:

keatuki wonder, premier, VPF – 191.

Phule Surekha:

Developed by selection from cv. Jama improved. Suitable for cultivation in Kharif, rabi and spring summer season. Pods are 9 -10 cm long, flat and light green in colour. Recommended for Kolhapur, Satara, and Sangli and Pune district of Maharashtra. Resistant to anthracnose, leaf crinkle, yellow mosaic and wilt disease. Average release committee of Maharashtra, it is also notified by the central subcommittee on crop standards, notification and varietal release at the national level.

Cultivation of Cluster Bean or Guar (Cyamopsis tetragoloba)

Botanical Name:

Cyamopsis tetragoloba L. Taub

Family:

Leguminaceae.

Cluster bean is grown for green vegetables and dry pods, and also as forage crop and green maturing crop. It is cultivated in many parts of India; some of its varieties are used for extraction of gum.

Climate:

Cluster bean is essentially a warm season crop and grows well in summer as well as rainy season crop. It is very hardy crop and relatively resistant to drought.

Soil and its Preparation:

Cluster bean can be grown on all types of soil. But well drained sandy loam soil is best suited for it. The land should be well prepared by repeated ploughing.

Sowing of Seeds:

a) Time of Sowing:

Early crop is sown from February to March and the main crop is sown with advent of monsoon, at the end of June or in beginning of July in certain area. Sowing is extended to September-October.

b) Seed Rate:

The seed rate is 5 to 8 kg/ha.

c) Layout and Spacing:

In layout flat bed and spacing 60 X 30 cm is recommended by P.K.V. Akola.

Manuring:

30 to 50 cartload should be applied at the time of soil preparation. 25 kg N, 50 kg P and 50 kg K/ha should be applied ½ dose or full dose P and K should be applied at the sowing. Remaining half nitrogen applied one month after sowing.

Interculture operation:

2- 3 weeding are required for control of weeds. Spraying insecticides and fungicides from time to time for control of insect pest and diseases.

Irrigation:

Irrigation should be given in early sown cluster bean when it is necessary. There is no need for irrigation for rainy season crop.

Harvesting:

The green pods for vegetable purposes are harvested from the plant by twisting or by cutting, when they attain the marketable stage. Harvesting or picking of green pods is continued over a considerably long time because they continue arising as the plant grows.

Yield:

The yield of green pods varies from 40 to 60 q/ha.

Varieties:

Pusa Mausmi (suitable for growing only as rainy season crop), Pusa sadabahar and Pusa Naubahar (suitable) for growing as a summer crop and rainy season crop) etc.

Cultivation of Cowpea (*Vigna sinensis*)

Botanical Name:

Vigna sinensis L.

Family:

Leguminaceae

Cowpea is a Kharif legume and is grown through India for green pod, dry seed, fodder and green manure.

Climate:

Cowpea is a warm season crop and cannot stand cold weather. Warm and moist climate is favorable for this crop. It cannot tolerate heavy rainfall.

Soil and its Preparation:

Cowpea can be grown in all types of soil. But sandy and sandy loam soils are best suited for it. The soil should be rich in organic matter. The land is prepared by giving four to five ploughing.

Sowing of Seed:**a) Time of Sowing:**

The time of sowing varies according to type of crop.

i) Kharif crop: May – June.

Ii) Rabi crop: October- November.

iii) Spring crop: February – March.

b) Seed Rate:

The rate is 15 to 20 kg/ha.

c) Method of Sowing:

The seed is sown by dibbling method. In layout flat bed and spacing 40 X 30cm, or 65 X 45 cm is recommended by PKV, Akola.

Manuring:

30 to 50 cart load of FYM or compost should be applied at the time of sowing preparation. 25 kg N, 50 kg P, should be applied ½ dose N, full dose P and k should be applied at the sowing and remaining half nitrogen applied one month after sowing.

Intercultural Operation:

No weeding is given this crop. It covers the land very soon and kill the weeds by smoothing. In early stage of growth, one weeding has been found beneficial to this crop.

Irrigation:

No irrigation is given in rainy season crop. But if grown earlier, the crop is irrigated whenever it is required. About three to four irrigation may be given before rainfall.

Harvesting:

The compare becomes ready for harvesting in 90 to 105 days for short duration crop and 135 to 150 days for long duration crop. Frequently tender pods should be harvested before they become fibrous. The plant is harvested for fodder or is used as green manure, when they are bearing flowers.

Yield:

Cowpea yield about 50 to 80 quintal green pods/ ha.

Varieties:

Pusa phalguni (suitable for sowing as a summer season crop), Pusa Barsati (Suitable for sowing as only rainy season crop), Pusa Dofasli (suitable for sowing as a summer and rainy season crop), Pusa komal.

Insect and Pest Management of Legume Vegetables

1. Aphids:

It is most destructive pest of legumes vegetable. These are very small insect and infest the leaves, stem and pods and suck the cell sap. In case of severe infestation, the infested parts dry up and there may not be any pod formation.

Control Measures:

This insect can be controlled by spraying the crop with Dimethoate (Rogor 30 EC) or Oxidomition Methyl (Metasystox – 25 EC) or methyl Parathion (Metacid 50 EC) @ 1 ml or Phosphamidon (Demicron – 100 EC) @ 0.5 ml per liter of water when the infestation is noticed.

2. Pea Pod borer:

The insect bore into pods and feed on the grains inside.

Control Measures:

Spraying the crop with Fenthion 0.05 percent and Phosphamidon 0.03 % can control this pest.

3. Pea Weevil:

The crop may become infested either in the field or in storage or in both places. It lays eggs on the young pods and leaves bore the seeds where they complete their development.

Control Measures:

Dusting the crop at blossoming time with one percent rotten one or 10 percent DDT can control this pest.

4. Leaf Eating Caterpillar:

The caterpillars feed on leaves generally during morning and evening and defoliate the plant.

Control Measures:

Dusting the crop with BHC or DDT 10 % @ 8 to 10 kg per acre or spraying the crop with Endosulphan (Thiodane 35 EC) @ 2 ml per liter of water has been found effective in controlling this pest.

Disease Management of Legume Vegetable

1. Seedling Blight:

Pre- emergence and post emergence damping off of seedling occur and the fibrous root being infested.

Control Measures:

The seeds be treated with Cerasan or Arasan or Speragon @ 2.5 gm per kg of seed before sowing.

2. Powdery Mildew:

Small, white, circular powdery spots appear on the upper surface of leaves which gradually cover the entire leaves, stems, petiole and the pods. They increased gradually resulting in the death of the leaves and fruits.

Control Measures:

Dusting the crop with sulfur @ 11.25 kg acer or spraying Wettable sulfur @ 2 kg %, starting just on the appearance of the disease have been found effective in controlling this disease.

3. Downey Mildew:

Symptoms are downy growth on lower surface of leaves which is white first, but later changes to violet and appears black in advance stage. Brown lesions are formed on pods in case of severe infector.

Control Measures:

This disease can effectively be controlled by spraying the crop copper Oxychloride (Blitox, Phytolan tc) @ 5 gm per liter of water.

4. Wilt:

Symptoms are premature yellowing and whitening of leaves and finally the diving of the entire plant. Plant may continue to wilt in patches in quick succession till the maturity period. The diseases affected plants can easily be pulled out. The diseases cause considerable damage when the crop is sown early and in light soil.

Control Measures:

1. Early sowing should be avoided in badly infested areas.
2. The seed should be treated with cerasn @ 2.5 gm or captan @ 2.3 gm per kg of seed.

ole Crops

There are the crop which belongs to the genus Brassica of the cruciferaceae family. They are essentially cold weather, hardy crops and thrive best in cool weather, hardy crops and thrive best in cool moist season.

They are generally grown throughout India during winter season. In summer season they are grown in parts of Bihar, Madhya Pradesh, Himachal Pradesh, Maharashtra and Nilgiri Hills in Madras. According to Choudhary (1967) all crops of this group have developed from wild cliff cabbage, also known as Cole worts, from which the name 'Cole' has been given. These crops if grown under warm conditions, the produce becomes pungent due to development of a glucoside called Sinigrin containing S-cyanoglucosin.

Though the crops of hills groups are unlike in vegetative character, yet they are similar in floral character and they can be readily crossed. The cultural requirement for all the crops are almost similar and many of the same diseases and insect attack there all.

Cole crop as a Group Includes:

- 1. Cauliflower:** *Brassica oleracea* var. *botrytis*.
- 2. White Cabbage:** *B. Oleracea* var. *capitata*.
- 3. Khol Rabi:** *B. Caulorapa*.
- 4. Sprouting Broccoli:** *B. Oleracea* var. *italica*.
- 5. Brussels Sprout:** *B. oleracea* var. *gemmifera*.
- 6. Chinese Cabbage:** *B. Chinesis*.
- 7. Kale:** *B. oleracea* var. *acephala*.

These crops are produced by transplanting and they do not set seeds in the plains except the early varieties of cauliflower.

Cultivation of Cauliflower (*Brassica oleracea*)

Botanical Name:

Brassica oleracea var. *botrytis* L.

Family:

Cruciferae.

Botany:

The edible portion of cauliflower is the site curd like mass composed of a close aggregation of abortive flower developed on thick bunches of the inflorescence. This edible portion is called 'Curd' surrounded by leaves which are longer but narrower than those of cabbage. According to the season of the growth, all types are divided into:

- a) Early season types and
- b) Late season types.

Late cauliflower types are grown for a longer period, have a larger plant and the leaves bend over the heads naturally and branch the curd while early cauliflower needs to have the leaves tied near maturity to branch.

Climatic Requirements:

Proper climatic conditions are almost essential according to the variety of the cauliflower. Cauliflower thrives best in a cool and moist climate. Cauliflower withstands so low temperature or so much heat as cabbage does. Dry weather and low humidity are not suited for it. For good seed germination, temperature of 50 to 70 °F is required. High temperatures produce poor quality curds viz Ricey, leafy, fuzzy, loose and yellow curds. Temperature below the optimum during growing period delays maturity and undersized, small unmarketable heads or buttons may be formed. Dry hot weather may give rise to small hard heads.

Soil and its Preparation:

It can be grown on a wide range of soil provided they are rich in nutrients and have adequate soil moisture, possess a good drainage and also contain plenty of organic matter. Sandy loam soil are preferred for early crops, while loam and clay loams for late crops. Cauliflower grows best on a neutral to slightly acid soils i.e. at pH 6.0 to 7.0. If the soil is below pH 5.5 liming at the rate of 5 to 10 quintal per ha should be done for successful raising of cauliflower. Higher pH than 7.0 decrease the availability of boron. The soil must be thoroughly prepared to make it loose and friable, and retentive of moisture. Basic organic manures should be applied during the field preparation. It requires better prepared soil than cabbage. That is why it has been humorously referred to as 'a cabbage with a college education.' 1 to 2 corrosive ploughing by a soil turning plough followed by 3 to 4 ploughing with desi plough are enough for it.

Seed rate and Time of Sowing:

The seed are sown in nursery bed in May- June for early, July – August for mid season (main crop) and September – October for late varieties. In cauliflower seed rate for early crop is 600 to 750 gm and for late crop 400 to 500 gm /Ha

Layout and Spacing:

Ridges and furrow type of layout is used for crop. Before that seedlings are prepared in nursery bed (Raised bed) and transplanted in main field after 3 – 4 weeks. Spacing for early crop is 45 X 45 cm and late crop it is 60X 60 cm.

Cultivars or Varieties:

In cauliflower various varieties are grown. They are season bound. Therefore, almost care should be taken while sowing the seeds. There are early mid season and late varieties, according to maturity in particular season.

Varieties suitable to be grown in rainy season (June – July) are Pusa Katki, Early Kunwari. The curds are available in September- October. Aghani matures in Nov, Pusa in December and maghi in January. Early cultivars produce short plant. The leaves are bluish green and produce small to medium and loose curd. Pusa Deepali is also early and curds mature in November. Curds are white. Pusa synthetic is another variety suitable for planting in September and curds are available in December – January.

The late cultivator is snow ball which can be planted in October – November and curds are available in February- March. The curds are white, compact and therefore yield is more. They also fetch more price in the market for all these varieties optimum temperature ranges for curd initiation and development varieties.

Manures and Fertilizer:

For best result 15 to 20 tons of FYM or compost should be incorporated into the soil about 4 weeks before transplanting. In cauliflower 100 kg N, 50 kg P₂O₅ /ha, should be given.

Irrigation:

It may be given to the crop every 5 – 6 days to the early planting and 10- 15 days for late crop. At the time of head formation, there should be enough moisture in the field, so irrigate at this time and when cauliflower is raised late in the season, it should be watered closely.

Interculture Operation:

Shallow frequent cultivation should be given in the cauliflower field by khurpi or hoe to kill young weeds and provide soil mulch. Avoid deep cultivation, for it may destroy the plant roots located top 3 to 6 meter in the soil. Weeding should be started as soon as plants are set in the field. Four to five weeks after transplanting, the plants should be slightly earthen up in the field.

Harvesting:

Cauliflower should be harvested when the head has developed the proper size and is at right stage of maturity. The head should be compact but it should not be broken into segments. The plant is cut off well below the head so that the stub has left the head from damaging during transporting to the market. The plants are cut as and when curds are well developed. As curds do not develop uniformly, so only those which are full developed are removed but not over matured ones after inspecting the whole field each second or third day. Harvesting the field during morning or evening so that the produce may be kept cool for the market.

Yield:

In case of early cauliflower crop 200 to 250 quintal / ha yield is obtained. While in case of later crop it is 250 to 300 quintal / ha.

Types of Cauliflower Curds

a) Ricey:

Due to the elongation of the peduncles bearing the individual flower buds, the curd become granular and loose. It is very undeniable and is formed during warm weather at harvest time, and when it is slightly over- matured. Good seed and culture under favorable condition may check Ricey heads.

b) Fuzzy:

It is velvety in appearance, and formed when the flower pedicels elongate. This type of curd may result due to hereditary factor, or when plants are grown under unfavorable condition and heads are over matured.

c) Leafy:

Curd becomes leafy due to the growing of small green leaves between the curd segments. They may appear in poor and sometimes in good strains also when growing conditions are unfavorable.

d) Perfect Curd:

In the type of lead or curd parts are well arranged to one another so that it looks almost homogeneous. This is found in young or partly developed heads. As soon as the segmentation begins to take place, the curd reaches full developed and maturity and is ready for the market.

Blanching in Cauliflower

According to Gupta (1947), Mehl (1959) and Srivastava (1960), blanching is an essential operation to protect the heads from sunburning and yellowing, so that they may not lose part of the flavor and attractive appearances. This operation may be done by drawing and tying of the leaves when the heads are formed. If it is not possible, place a cauliflower leaf over the heads to protect it from the sun. This will induce milky white colour to the curd which may be sold at premium in the market.

According to the variety heads are ready to harvest in 60 – 70 days (Early), 90- 100 days (Medium), and 110- 120 days (Late) after transplanting. Blanching should be done only when the head has grown fully, and leaves should not be left tied over for more than 4- 5 days. But these periods are increased to one week in cold weather and reduce to 2 – 3 days during the hot weather. This process is called as blanching.

Disorders in Cauliflower

Various disorders are observed in Cole crops. These are physiological disorders. Some disorders are also caused by climate and injury to growing bud.

Buttoning:

The buttoning means production of small button like curds. This is due to sudden check in the vegetative growth. The reasons attributed are over aged seedlings, poor nitrogen supply, planting early varieties in late season and change in temperature.

Riciness:

Premature initiation of floral bud is characterized by riciness. Such curds are poor in quality for marketing. Riciness results due to unfavorable temperature for a given variety.

Browning:

This is caused due to deficiency of boron in soil. In serious cases stem becomes hollow with water soaked tissues. In advanced stage pinkish or brown area develops on surface of the curd. This can be controlled by applying sodium borax 20 kg/ ha or spraying of boron 0.25 to 0.50 % solution.

Whiptail:

This is caused due to molybdenum deficiency. Young plants turn white particularly along the leaf margins and also become cupped and whiter and dies. Apply molybdenum @ 1 kg/ ha to control the malady. Leaf blades do not develop properly, only midrib develops.

Blindness:

Blind cauliflower plants are those without terminal buds. The leaves which develop are large, thick leathery and dark green. Blindness is supposed to be due to temperature or injury due to cultural operation, insect and pest disease.

Cultivation of Cabbage (*Brassica oleracea*)

Botanical Name:

Brassica oleracea Var. *Capitata* f. *alba*.

Family:

Cruciferaeae.

The peculiar flavor in the head is due to the glucocide Sinigrin, which carries sulphur also. The open green leaves are more nutritious than cabbage head.

Botany:

The word cabbage is derived from the French word 'coboche', meaning head. A cabbage head is made up of numerous thick, overlapping smooth leaves which cover smooth terminal bud. Sometimes small heads of 5 or 7.5 cm. In diameter are formed, which are known as 'cabbage sprouts' having no commercial importance. Normally it is biennial but it is grown in India as annual crop.

Season and Climate:

It grows best in cool moist climate and is very hardy to frost. In areas with comparatively dry atmospheres, its leaves tend to be more distinctly petiole than in the more humid areas. In hot dry atmosphere, its quality becomes poor and much of its delicate flavor is lost. Its germination is best at a soil temperature of about 55 °F to 60 °F. Temperatures below this and above this are not suited for it. Well hardened seedlings can tolerate temperature of 20 °F to 25 °F. It is grown mainly as rabi crop during winter. But in and around Nasik (Maharashtra), Ootacamond (Madras), and in semi parts of Kerala, it is grown as kharif crop also.

Soil and its Preparation:

It can be grown almost in all types of soil ranging from sand to heavy soils. But small quick growing cabbage varieties do well in sandy soils, while large and late maturing varieties in heavy soils. While large and late maturing varieties in heavy soils. Soils intended for cabbage growing, should have good drainage. Acid soils are not good for cabbage. The best PH range for cabbage is between pH 5.5 to 6.5. Lime may be added in acid soil to make it neutral or alkaline for growing good crop of cabbage. Land is prepared by ploughing it 3 to 4 times. The first ploughing should be done by soil turning plough, and the bulky organic manures should be spread in the field. Then the land should be ploughing and leveling the land, beds of suitable size and irrigation channels are made.

Seed Rate & Time of Sowing:

Cabbage is grown mainly as Rabi crop during winter (Sept.-Oct.), But around Nasik (Maharashtra) it is grown as kharif crop also. In cabbage for early crop (September) 500 gm & for late crop 373 gm. seed is required.

Layout & Spacing:

Ridges & furrow type of layout is used for crop. Before that seedlings are prepared in nursery bed (Raised bed) & transplanted in main field after 3-4 weeks. Spacing for early crop is 45x45 cm and late crop it is 60x60 cm.

Manures & Fertilizers:

150 kg N, 50 kg P₂O₅ should be applied per hectare.

Irrigation:

Irrigation at the time of transplanting is essential. Steady supply of moisture is necessary for good growth and development. Interval between two irrigations depends upon climate, soil and plant growth. In winter season irrigation at an interval of 8-10 days is sufficient.

Harvesting and Yield:

In cabbage harvesting is done depending on the maturity of the head and demand in market. Normally harvesting is done when heads are firm. If prices are high in the market harvesting is done earlier when heads are small and loose. Heads are cut with a knife with little stalk and some leaves. Proper grading is followed before heads are sent to market. The yield of cabbage variety from 15 to 25 tons per hectare. The yield of cabbage depends upon the variety, growing season and management practices. Hybrid cabbage yields up to 50 tonnes per hectare. The yield of early varieties ranges between 12 to 15 tonnes/ha. The yield of late season varieties is about 20 to 25 tonnes/ha.

Cultivars or Varieties of Cabbage

Cabbage cultivars are classified on the basis of colour of head and maturity. In India white cabbage is important. These cabbages are available in pointed, round and flat shapes, of these round cabbages are preferred. Common cabbage varieties grown along with their characteristics are as below.

Early varieties:

Golden Acre:

It has compact and round head, plant is small and compact with few outer leaves with short stem. Interior position of head is white. Average weight of head is 1 to 2 kg. It is ready for harvest in 70-80 days in proper season.

Pride of India:

It is one week later than Golden Acre. Head weight is 1 to 2 kg.

Copenhagen Market:

It is later by one week than Pride of India. Ready for harvest in 90-100 days and head weighs 1.5 to 3.00 kg. Head is compact.

Late Varieties:

In late varieties flat shaped head varieties are grown. Pusa Drumhead is a late selection. Heads are large flat and somewhat loose. Average head weight is 1.5 to 3 kg. It is a heavy yielder.

Hybrid Cultivars:

Hybrid cultivars are introduced which are more uniform in maturity, size and shape of head than common cultivars. Hybrid cultivars can be grown at higher temperatures. Hari Rani is one of the best hybrids. Now a day's cabbage is grown in different seasons. It has become possible because of hybrid cultivars which can tolerate high temperature. Various private seed companies are marketing different hybrids viz. Kaveri, Hardam, Nakshatra etc.

Cultivation of Knol – Knol (Brassica caulorapa)

Botanical Name:

Brassica caulorapa L.

Family:

Cruciferaeae.

Khol Rabi which is the German name for cabbage turnip resembles an above ground turnip. It is also called knol-knol. The fleshy edible portion is an enlargement of the stem, which develops entirely above ground and is used as a vegetable. It is an excellent vegetable if used at its early stage before it becomes tough and fibrous. The edible portion is globular to a slightly flattened stem.

Climate:

Knol- Knol is a cool season crop and thrives in a relatively cool moist climate. It can withstand extreme cold and frost better than other cool season crops. Seeds of knol- knol germinate well at 15 °C to 30 °C.

Soil and its Preparation:

Knol- Knol is successfully grown on all types of soil. But the sandy loam and clay loam soils are best suited for knol-knol crops.

The land should be well prepared and of good tilth for sowing of Knol- Knol, 5- 6 ploughings should be given to prepare a good tilth. After ploughing, planking is essential to make soil pulverised and leveled. Harmful weeds and stubbles of previous crop should be removed before the transplanting of this crop.

Aldrin or Heftaf or Heptachlor 5 % or BEC 10 % should be applied @ 10 – 15 kg per hectare with the last preparatory tillage to check the infestation of termites and cutworm.

Sowing of Seed:**a) Time of Sowing:**

Knol-knol is normally propagated by seeds. It can also be propagated by cuttings such as tip cutting and leaf cutting. The seeds are sown in lots from the end of August to the end of November for a continuous crop. Time of sowing of seeds varies according to the variety viz.

Variety	Time of sowing seed in Nursery	Time of transplanting in the main field
Early	August	September
Main	September	October
Late	October	November

b) Seed rate:

1 to 1.5 kg seed is sown in nursery to cover one hectare of land.

Transplanting:

3 to 4 week old seedlings become ready for transplanting. Ridges and furrow type of layout is used. The seedling is transplanted at a distance of about 30 cm. from row to row and 20 cm. from plant to plant. Closer spacing of this vegetable increases the yield. After transplanting, the seedlings should be watered and protection may be given from sun rays for 4 – 5 days.

Manures and Fertilizer:

Mix 15 -20 tones of FYM to soil at the time its preparation. At the time of transplanting add 75 kg n, P and 50 kg K/ ha, second dose of 50 kg N should be given 1 ½ months after transplanting.

Interculture Operation:

Hoeing and weeding should be done after each irrigation when the soil comes in working condition. The plant should be earthen up after 25 – 30 days of transplantation.

Irrigation:

As stated the seedlings, after transplanting, should be watered for four to five days. Irrigation should be given an interval of 6 to 7 days. But the field may be irrigated when soil is deficient in moisture.

Harvesting:

Knol-knol should be harvested when swollen stem reach a diameter of 5 to 7 cm and before it becomes tough and woody. The plants are pulled out from the land. It is generally marketed after removing both leaves and roots.

Yield:

250- 300 quintal yield per hectare is obtained.

Varieties:

As recommended by IARI, New Delhi.

Early: earliest, white Vienna etc.

Late: Purple, Vienna, etc.

Insect Pest Management of Cole Crops

1. Mustard Saw- Fly:

Scientific Name:

Athalia proxima Klug.

The blackish caterpillar that attacks all cole crops and feed on leaves of young seedling in the early stages. The infested leaves become curl and falls on the ground when touched.

Control Measures:

i) Hand picking of caterpillar, if infestation is light and number of attacked plant is small is the best method of controlling the larvae.

Ii) Dusting the plants with BHC or DDT10 % of sevin @ 10 kg % acer or spraying the same @ 5 gm per liter of water is also effective to control this insect.

2. Mustard Aphids:

Scientific Name:

Lipaphis erysimi Davis.

A greenish white small insect that attack cabbage and other cole groups when there is a cloudy weather. They suck the sap from the plants. The affected leaves get curled and plants wither and die's.

Control Measures:

Spray Endosulphan 35 EC @ 0.5 or Phosphamidon 85 @ P 0.02 % or Dimathoate 30 EC @ 0.5 % or Methyldemeton 25 EC @ 0.02 % or Thiomethon 25 EC @ 0.025%.

3. Diamond Black Moth:**Scientific Name:**

Plutella maculipennis curt.

A small slender pale green caterpillar that feeds on leaves and make holes in them. The infested crops become unsuitable for marketing.

Control Measures:

This pest can effectively controlled by spraying the crops with Malathion (Cythion- 50 EC, Folithion – 1000) / 2 ml per liter of water or Dichlorodane (Nuvan – 1000, Vapona Agrosan etc) or Monocrotophos (Monocil, Nuracron, Suphos etc) @ 1 ml per liter of water.

4. Cabbage Borer:**Scientific Name:**

Hellula undalis F.

A whitish brown caterpillar that mines the leaves and feed on shoot. Later it burrows into the stem or cabbage. The affected plants wilt, remain dwarf or may even die.

Control Measures:

i) Removal of caterpillar when the plant is young is the best method for the control of larvae.

ii) This pest can effectively be controlled by spraying the crop with Phosphamidon (Demicron- 100 E. C sumison, demeton etc.

5. Cabbage Fly:**Scientific Name:**

Chotochila Brassica (hylemya)

The maggots first attack the young root lets and then burrow into the main root. The affected plants become yellow, wilt and sometime they may die.

Control Measures:

Soil application of aldrin or heptachlor 5 % @ 15 kg percent acre or heptaf 5% @ 10 kg acre at the time of planting have been found effective in controlling this pest.

6. Painted Bug:

Scientific Name:

Bagda cruciferum kirk.

The adult and nymph of this pest suck the sap from the plants and related their normal growth.

Control Measures:

A greenish white small insect that attack cabbage and other cole groups when there is a cloudy weather. They suck the sap from the plants. The affected leaves get curled and plants wither and die's.

Diseases Management of Cole Crops

1. Leaf Spot and Bight:

Small dark coloured spots which spread rapidly to form circular lesions and concentric rings of dark conidiophores' appear in humid weather due to which cauliflower develops brown discoloration and cabbage after harvests acquires black mouldy appearances.

Control Measures:

i) Hot water treatment of seeds at 50 0 C for 30 minute helps to check the spread of this diseases.

ii) Spraying the crop with Zineb (Dithane Z – 78) or Mancozeb (Dithane M – 45) @ 2.5 gm per liter of water or copper oxychloride (Blitox, Blucopper, Phytolap, cupravit etc.) @ 5 gm per liter of water at an interval of 10- 15 days according to incidence of diseases is very effective to control this disease.

2. Black Rot:

The margin of disease affected leaves turn yellow vein becomes dark and vascular discoloration takes place in the main stem. Cabbage and cauliflower heads are discolored scheduling may die when attacked in young stage.

Control Measures:

The disease can be controlled by seed treatment with hot water at 50 0 C or one % mercuric chloride solution for half an hour before sowing and adopting crop rotation with non-cruciferaceous crops for three years.

3. Club Rot:

The rots of affected plat enlarge to from clubs secondary invasion by soft root bacteria follows forming material toxic to plant. The foliage of affected plants with on sunny days and recovers towards evening.

Control Measures:

The seedlings at transplanting time should be treated with mercuric chloride @ 125 c.c (about 113.4 gm).

4. Damping Off:

It is a common problem in the nursery bed of vegetable crops. The affected plants fall down and die due to shrinking of cortical tissue of hypocotyls.

Control Measures:

Drenching the nursery bed with formaldehyde or brassicol about 3 -4 weeks prior to sowing, seed treatment with mercurial fungicide @ 2 gm per kg of seed, avoiding thick sowing and spraying the seedlings with Dithane M - 45 or Dithane Z- 78 @ 2.5 gm per liter of water are the common remedial measures.

Seed Production of Cauliflower

It is a cross pollination vegetable crop. Isolation distance should be 1200 meters. Seeds of early variety can be produce in plains of India having cool climate. But late varieties can be produce only in hilly region. Seeds of early varieties are produce mainly in Uttar Pradesh and Bihar states of our country. Seeds of alter varieties was imported from other varieties previously, due to difficulties in seed production, but now a days by using new techniques, the seeds of late varieties i.e. snow ball are produce in Kullu valley, Kashmir, Darjiling, Nilgiri Hills of our country as late varieties require very cold climate during its seed production.

eed Production of Cabbage and Knol- Knol

These are cross pollinated crop. They, required very cold climate during its seed production, produce only in Kullu and Kashmir valley of Himalaya of our country.

Cultivation of Reddish (Raphanus Sativus)

Botanical Name:

Raphanus Sativus L.

Family:

Cruciferae.

Among the root crops carrot and radish are the important crops grown in the state. Radish is a popular vegetable in both tropical and temperature regions. Radish is a good source of vitamin C contains 15 – 40 mg per 100g of of edible portion and supplies varieties of mineral. The leaves of radish are a good source of extraction of proteins on a commercial scale.

Radish is grown for its tender tuberous roots which are eaten raw salad or 8cooked as vegetable. The leaves of radish are also cooked as vegetable. Root, leaves, flower and pods are active against gram- positive bacteria. The roots are said to be useful in ruminary complains and piles. Radish is annual or biennial depending on purpose for which it is sown.

Climate:

Radish is best adapted to cool or moderate climate. Tropical varieties can adopt high temperature but to develop good flavor texture lower temperature between 15- 25 0 C is required initially 20

and 23 °C temperature is required. During hot weather the roots become tough and pungent before reaching edible size. However, pungency is also a varietal character.

Soil:

Radish can be grown in all types of soil but best results are obtained on light, friable loam soil that contains humus. For early crop sandy or sandy loam soils are preferred. Usually heavy soils produce rough, misshapen roots with a number of small lateral roots and such soils should be avoided. High yields are obtained on heavy soils.

The soil should be thoroughly prepared and no clods or under decomposed manures should be left in the soil. The clods interfere in the development of the roots. So bring soil to a fine tilth before sowing is undertaken.

Sowing and Seed Treatment:

The radish crop can be sown on ridges and furrows or on flat beds too. The distance between ridges and furrows or on flat beds depends upon variety.

The temperate varieties and early varieties are sown at a closer spacing. Radish can be sown from July- January. However, all varieties of radish may not perform well in all the months. During hot weather or in rainy season tropical varieties can be grown. The tropical varieties sown after November wilt bolt. The radishes are also sown in Kharif season but the root gives good quality is inferior. Radish is sown on ridges and furrows opened at 45 cm distance and seeds dibbled at 8 to 10 cm in row. Surface sowing or sowing at depth of 2- 3 cm is advisable for proper germination. On flat beds sowing at 15 X 15 cm can be followed. 8 -10 kg of seed required for one hectare area.

Manuring and Fertilizer:

To obtain good yield of excellent quality roots, judicious application of manures and fertilizers is recommended. Radish is a short duration crop. Do not apply fresh and uncomposted manure in the field as it results in forking and misshapen roots. Application of 50 kg Nitrogen, 25 kg Phosphorus is recommended by PKV, Akola. Half of the dose of nitrogen applied is 20- 30 days after sowing.

Irrigation:

Radish requires steady and continuous supply of soil moisture for better root growth. This also helps in production of tender and attractive root growth. At the time of sowing there should be sufficient moisture. So irrigate field before sowing. In dry weather frequent irrigation is necessary. Water stress makes roots rough and pungent. In winter season irrigation at interval of 6-8 days are given.

Interculture:

As it is sown at closer spacing, weeding is difficult hence soil should be well prepared. After sowing regular stage of crop growth is essential. In radish roots there is tendency to bulge out of the soil therefore cover roots by earthing up.

Harvesting:

Depending upon varieties roots become ready for harvesting in 40- 45 days after sowing. Early rapid maturing varieties become ready for harvesting even at 25- 30 days of sowing. Harvesting radish at proper stage of maturity. Delay in harvesting results in pithiness and bitterness in radish roots.

Give light irrigation before harvesting to facilitate easy pulling of roots, the roots along with leaves are easy pulling of roots. The roots along with leaves are washed cleaned and tied in bunches.

Yield:

Average yield of radish varies from 100 to 180 quintals per hectare. However, some cultivars also yield. 400- 500 quintal per hectare. Yield of temperature varieties is less.

Varieties of Reddish

Large numbers of cultivars are cultivated. There are two types of cultivars. Temperature of European and Tropical or Asiatic. European types growth in the region does not flower.

Ganesh Synthetic:

Ganesh synthetic is developed at regional fruit research station, ganeshkhind, pune. Roots are attractive white in colour, mild pungent. Ready for harvest within 30- 40 days. Yield with and without leaves is 500- 800 and 300-600 quintals per hectare respectively.

Pusa Desi:

it is tropical, subtropical, suitable for sowing middle of august to October. The roots are pure white, 30- 35 cm long tapering with green stem and pungent and heavy yielded.

Pusa Chetki:

It can tolerate high temperature; it can be sown from middle of august. This cultivar sets seeds under tropical condition. Roots are medium large and stumpy, pure white, tender and smooth and mildly pungent. It becomes ready for harvesting in 40- 45 days after sowing.

Japanese White:

It is excellent variety so far as root quality is concerned. The roots are cylindrical, 22- 25 cm in length and 5 cm in diameter. The skin is snow white, fresh crisp, solid and less pungent. It matures in 45- 50 days. It is suitable for growing between November to December.

Kalyanpur Selection:

It is a tropical variety and heavy yielder. The tips are more in proportion than roots. It is suitable for growing in winter season. Sudden drop in temperature induces bolting.

Besides these varieties, there are varieties of good root qualities like Pusa reshmi, Pusa Himani. The variety Janupurin Giant is also a tropical variety and heavy yielder. Radish is very specific in its climatic requirements especially temperature. Therefore, it is very important to select right cultivar for growing in the season.

Disease and Pest Management in Reddish**Diseases:**

Radish is a short duration crop. Therefore, prompt care has to be taken to control diseases and pest. Alternate blight white rust is important diseases. Alternaria blight attack on leaves seen in the form of small, yellowish, slightly raised lesions. It is more in rainy season. Spray crop with copper- oxychloride to control disease. The disease is not so important in winter season in our area.

Pests:

1. Mustard Saw Fly:

Mustard saw fly is most destructive is most destructive insect of radish in our region. If not controlled in time causes serious damages to the crop. It attacks the crop on leaves. The grub feeds on the leaves and pods. In severe cases complete defoliation occurs. Spray with Malathion to control disease.

2. Aphids:

Aphids are most serious pest particularly in rainy season. Under severe infestation plants are completely devitalized. Leaves and shoots curl up, become yellowish and finally die. Spraying with Malathion 0.05 per cent check the attack of aphids.

Cultivation of Carrot (Daucus Caroto)

Botanical Name:

Daucus caroto L.

Family:

Umbellifera.

Carrot is most important root crop in curries and made into pickles and sweet meats, 'Gajar Halwa' is a delicious dish prepared from carrot. Carrot juice is also used. Carrot is rich source of carotene a precursor of vitamin A. the tops of carrots can be used as source of essential oil. Carrot has medicinal properties also. Carrot is annual or biennial herb with erect to much branched stem. The edible portion or carrot root is an enlarged fleshy tap root. It consists primarily of phloem or cortex and minimum of core. Carrot plants bear compound umbel. It is the umbel or umbel of the first order or primarily umbel that flower first. In succession, secondary and tertiary umbels are formed at certain interval.

Climate:

Carrot is grown in winter i.e. from October – November. Very high or low temperature is not conducive for root development. Carrots produce largest roots with good colour are sown by the end of October to November to obtain high yields, with good quality roots.

Soil:

It can be grown on all types of soil but best growth is obtained in a deep, loose, loamy soil. For early crop sandy loam soil preferred but for large yield silt-loam is desirable. In well drained, deep soil, long smooth slender roots are formed. The carrots grown on heavy soil are rough and coarse. Maximum yield is expected at soil PH 6.5.

Since roots develop the soil. It is necessary to prepare soil to a fine tilth. If soil is not thoroughly prepared and if it contains soil clods or underdecomposed organic matter, good quality and well shaped roots cannot be produced.

Sowing:

Carrots are grown from direct seedling,. The seeds are sown either by broadcasting or drilling in lines. To facilitate even distribution seeds are mixed with fine sand. Seeds can be sown on flat bed at wider spacing of 15-25 cm or on both sides of ridge, while sowing soil should have sufficient moisture. After sowing, give light irrigation with due care to avoid flooding of seeds. Follow shallow sowing for better seed germination. Seed soaking in water for 12-24 hours before

sowing gives good germination. Follow sowing at 15 X 15 cm distance in flat bed. In case of ridges and furrow layout 45 X 15 cm spacing is kept. On an average 6 -8 kg seed is required for a hectare.

Manures and Fertilizer:

30 to 50 cartload has a fairly extensive and widely drenched root system. Carrot can absorb water from deeper layer. Carrot grows on well retentive of soil moisture needs less water. It needs frequent irrigation during germination and vegetative growth which gives more yield. Irrigation frequency should be reduced on month before the due date of harvesting. Irrigation is almost stopped 15 – 20 days before harvesting which helps increasing sweetness. Excessive irrigation induces adventitious roots can carrot which impact root quality.

Interculture:

Prompt weeding in early period of one half month after sowing is essential as the cross sown closely. Later on it is difficult to weed. In this regard clean cultivation before sowing is important. Weed competition causes heavy loss of top and impair root quality. Manual weeding not only control not only controls the weeds but improves aeration resolution in better root growth. Earthing up is also practiced which control the weeds and also covers the exposed roots to prevent discoloration of roots.

Harvesting and Yield:

Harvesting of carrots at proper stage of maturity is essential to fetch good price in the market. Delay in harvesting deteriorates the quality of the roots and becomes unfit for consumption. Every variety has certain period for harvesting. Delay in harvesting results in splitting of roots. Formation of adventitious roots, increase in dry matter sucrose, carotene and bolting.

Harvesting is done manually. Give light irrigation before harvesting to facilitate easy pulling of roots without damage. In heavy soil, roots are removed from soil by digging. After harvesting roots are washed, cleaned, graded and tied in bunches. On an average tropical carrot up to 150-250 quintals/ hectare.

Factors Affecting Colour, Shape and Size of Carrot

Colour of the carrot is due to presence of pigment. The orange colour is due to carotenes and yellow due to anthocyanin. The variation colour is due to accumulation of carotenoids in varying degree. The colour development depends on cultivar, growing season and age of the root. A temperature of 15 to 21.1 °C is best for colour development. Wet weather proceeding to harvesting results in low carotene. Carotene synthesis is more in bright sunshine. The shape and size of carrot roots are influenced by soil type, temperature etc. branching in carrot due to hereditary, and presence of undercomposed organic matter or injury to the tap root system or check in heavy soil at lower temperature. Roots are longer and slender at 13 to 20 °C than at higher temperature. High temperature and irregular results in rough root surface.

Cultivars/ Varieties of Carrot

Varieties of carrot are classified on the basis of shape of root and temperature response to flowering. There are carrot varieties such as long rooted, half- long rooted and short stump rooted. On the basis of temperature requirement, carrots are classified as temperate or European and tropical or Asiatic. The temperate varieties are biennial and require low temperature (4 – 8- 10 °C) treatment for flowering to colour. These do not produce seeds in places e.g. Nantes, Danvers Half Long etc. the tropical cultivars do not require any low temperature treatment for flowering. They produce flowers and seed in places of high temperature.

Pusa Kesar:

This is a selection from a cross between Local Red and Nantes Half Long. This top is medium. The roots are long and red in colour with small core. It contains high (38 mg/ 100 kg) carotene. The percentage of forked roots is less and it is late bolter. Pusa Kesar can tolerate high temperature than Nantes. It is recommended for this region.

Nantes:

It is European cultivars which can be grown for root production. The roots are half long, slim, well shaped with stump end. The colour is deep orange and it has fewer cores. It contains high carotene. It has good qualities of root hence recommended for the region

Disease and Pest Management in Carrot

Disease and pests are not very serious problem in carrot cultivation. Common diseases of top of carrots are Alternaria blight, leaf spot and powdery mildew.

Leaf Blight:

Due to leaf blight small dark brown to black spots with yellow edge appear on underneath the leaf and assume brown or purplish colour. Spraying or dusting with sulphur checks the disease.

Powdery Mildew:

Powdery mildew occurs on leaf stem and flowers. Powdery growth is observed underneath the leaf and assume brown or purplish colour. Spraying or dusting with sulphur checks the disease.

Cultivation of Garlic (*Allium sativum*)

Garlic:

Botanical Name: *Allium sativum* L.

Family:

Amarylidaceae.

Garlic is one of the important bulb crops. It is used as a spice or condiment through India. The compound bulb of garlic consists of several small bulblets or cloves. Garlic is generally cultivated in A. P, U.P, Madras and Gujarat.

Composition and Use:

Garlic has been considered as a rich source of Carbohydrates, proteins Phosphorus. Garlic contains allin. Its principle ingredient is diallyl disulfide. It contains volatile oil. Garlic helps indigestion. It reduces cholesterol in human blood.

Climate and Season:

It grows under a wide range of climatic conditions. However. It cannot stand too hot or cold weather. It prefers a moderate temperature in summer as well as in winter extremely hot and long days are not conducive are not conducive to proper bulb, short days are very favorable for the formation of bulbs. It can be grown well at elevations, of 1000 to 1300 meters above sea- level. In

India it is cultivated in Madras, Andhra Pradesh, Uttar Pradesh, Uttar Pradesh and Gujarat (Rajkot division).

Soil and Planting Time:

According to Rao and Purewal (1957) and Joshi (1961), garlic requires medium black to well drained loamy soils rich in humus, with fairly good content of potash. The crop raised on sandy or loose soil does not keep for long and the bulbs too are lighter in weight. In heavy soils, the bulbs produced are deformed, and during harvesting, many bulbs are broken and so they do not keep well in storage. They get discolored in badly drained soils.

Seed Rate and Time of Sowing:

In garlic seed rate is 315 to 500 cloves per ha. It is taken in rabi season and summer season. Crop is planted from August- November.

Layout and Spacing:

Ridges and furrow type of layout is followed. Spacing is given.

Improved Varieties:

Fawari, Rajli gaddi, G-41, Selection – 2, Selection – 10, Garlic 9 (*Allium Sativum*):

1. Godavari:

Developed by selection from Jamnagar collection and released in 1987. Bulbs are pink in colour and medium in size with 25- 30 cloves per bulb. It is tolerant to eriophyte mites. Duration is 130- 140 days. Duration is 130- 140 days. Average yield is 150 quintal per hectare.

2. Sweta:

Sweta is developed by selection from a local germplasm collected from Gujarat and released in 1987. Bulbs are medium in size with 20-25 cloves per bulb of white colour. Duration is 120- 130 days. Average yield is 130 quintals per hectare.

Manures and Fertilizer:

25 to 30 cart load of well rotten FYM or Compost applied at the time soil preparation per hectare. In garlic according to PKV, Akola recommendation 50 kg N, 50 Kg P₂O₅ and 50 kg K₂O should be applied. A side dressing of 50 kg N may be given one month after sowing.

Interculture Operation:

First Interculture is given with hand or khurpi one month after the sowing. Second weeding is given one month after the first weeding and hoeing. According to Joshi (1961). Hoeing, the crop just before the formation of bulbs about 2 ½ months from sowing) loosens the soil and helps in the setting of bigger and well filled bulbs. The crop should not be weeded out or hoed at a later stage because this may damage the stem and impair the keeping quality of sprouting and the cloves already formed.

Irrigation:

First irrigation is given after sowing and then field is irrigated after every 10 to 15 days till the season warms up. At this stage more frequent irrigation are required. There should be no scarcity of moisture in the growing season; otherwise the development of the bulbs will be checked. When

the bulbs are matured then irrigation should be given sparingly. The last irrigation should be given 2 -3 days before harvesting for making it easy without damaging the bulbs.

Harvesting:

Garlic is crop of 4 ½ to 5 months duration. When the leaves start turning yellowish or brownish and show signs of drying up (usually about a month or so for the emergence of seed stalks, the crop is ready for harvest. The plants are then pulled out or uprooted with a country plough and are tied into small bundles, which are then kept in the field or in the shade for 2- 3 days for curing and drying so that the bulbs become hard and their keeping quality is prolonged. The bulbs may be started by handling them on bamboo sticks or by keeping them on dry sand on the floor in a well ventilated room on dry floor.

Yield:

In garlic 50 to 70 quintals / ha yield is obtained.

Insect Pests Management in Garlic

1. Thrips:

This is surplus in garlic. It is a small sucking insect that feed on leaves and lacerate tissue.

Control Measures:

This pest can be controlled by spraying the crop with Dimethoate (Rogor 30 EC) or Methyl Parathion (Metacid – 50 EC) or Oxydemeton methyl (Metasytox – 25 EC) @ 1 ml per liter of water.

2. Onion Maggot:

This pest damage the crop by eating bulb and the subsequent decay of the affected portion frequently destroying them.

Control Measures:

Application consist of placing sand soaked with a cupful of kerosene in a bucket of dry sand at the base of the plants along the row is very effective for deterring the parent flies from depositing their eggs. This will also kill young maggots that might attempt to work through sand.

3. Black Onion Fly:

It feeds on onion, garlic and chews both in the field and in stores.

Control Measures:

Application consist of placing sand soaked with a cupful of kerosene in a bucket of dry sand at the base of the plants along the row is very effective for deterring the parent flies from depositing their eggs. This will also kill young maggots that might attempt to work through sand.

4. Garlic Mite:

It suck the sap from the plant and causes serious damage of this crop.

Control Measures:

i) The bulb should be soaked in solution of 150 ml Ekalinin 50 liters of water for 10 minutes and dried before planting.

ii) Spraying the crop with kelthane @ 5 ml liter of water can control this pest.

Disease Management in Garlic

1. White Rot of Garlic:

It is a common disease of garlic in storage. Dirty white spots appear in the fleshy cloves and this increases in size with the time of storage.

Control Measures:

i) The infested bulb should be sorted out and destroyed.

ii) The stored bulb should be turned frequently to check the spread of this disease.

2. Blast Garlic:

Dark brown spots with ash coloured centers appear on the leaves.

Control Measures:

Removal of the affected plants as soon as diseased plants are moisture and spraying the crop with Mancozeb (Dithane M – 45 @ 2.5 gm per liter of water) are the common remedial measures.

Cultivation of Potato (*Solanum tuberosum* L.)

Botanical Name:

Solanum tuberosum L.

Family:

Solanaceae.

The Potato (*Solanum tuberosum* L.) belongs to the family Solanaceae. There are many related species, but only the potato and a few others bear tubers.

Potatoes are an important staple food crop. Annual world production of potatoes surpasses that of all other vegetables and ranks with wheat and rice. Introduced into Europe in the 16th century by Spanish explorers, it is cultivated throughout the world including the Indo-Pakistan subcontinent. Potatoes are used alone and mixed with meat, fish, chicken, and vegetables, & in big cities and towns, potato chips are prepared and sold commercially. The potato is rich in starch and contains protein, minerals, & a fair amount of vitamins, particularly Vitamin - C.

Climate:

The potato has a wide range of seasonal adaptability. It is a cool season crop and is moderately frost - tolerant. Temperature during the growing season has long been recognized as one of the most important factors influencing yield. Young plants grow best at a temperature of 24°C; later growth is favored at 18°C. Tuber production reaches a maximum at 20°C, decreases with rise in

temperature, and at about 30°C tuber production stops entirely. Short days are beneficial for tuber production.

Soil:

The potato develops best on deep, fertile, sandy lo clay loams with good water retention capacity. Because the potato has a relatively weak, shallow root system, impermeable layers in the soil limit rooting depth, which restricts water availability to the plant in dry periods. Thus soil compaction can greatly reduce potato yields. Aeration of the soil has a great effect on the set and development of tubers.

Manures and Fertilizers:

Potato has a high nutrient requirement; a yield of 25 tones of tubers/ha depletes the soil of 119 kg of N, 50 kg of P and 225 kg of K. In light soils and places where organic manures are not easily available, green manuring is beneficial. Well rotted FYM at 30 t/ha should be incorporated into the soil three to four weeks before planting. Research on fertilizer requirements conducted in the Punjab and Sindh suggests that 100 kg N, 80-100 kg of P₂O₅, and 50 kg K₂O per hectare are sufficient- to obtain good yields.

Spacing:

In hilly areas, potatoes are spaced at 60 x 30 cm and the tubers are planted in furrows. In the plains, however, they are planted on ridges spaced 30 cm apart; the distance between ridges is kept at 45-60 cm.

Seed Rate:

Potatoes are propagated by tubers, planted either whole or cut into pieces. To obtain maximum yields, healthy, disease-free tubers, free from mixture of other varieties, should be used. Seed rate depends on tuber size; 800-1000 kg/ha is generally recommended. Choudhary et al (1990) conducted a trial comparing improved seed of the Swat and Kaghan varieties brought from the Kaiani Integrated Development Project with seed from the local market. Potatoes grown from the improved seed gave better soil coverage, had a lower incidence of virus attack and higher tuber weight and gave higher yields than those from the locally purchased unimproved seed.

Irrigation:

Potatoes need frequent irrigation. The first irrigation should be given immediately after sowing, and thereafter at one week intervals. After tuber formation, the frequency of irrigation is decreased. Care must be taken while irrigating that the water does not reach the top of the ridge. Irrigation should be stopped a few days before harvesting.

Harvesting:

The size of the tuber increases until the vines become dry. In developed countries, potatoes are harvested by mechanical harvesters. In Pakistan, they are manually harvested by digging up the ridges with a spade. Care must be taken that tubers are not injured during the process. After harvesting, the crop should be kept in the shade.

Yield:

The average yield of potato is 20-25 tonnes/ha, however it depends upon the variety, cultural practices, and location.

Varieties of Potato

Commercial potato varieties released by C.P.R.L Shimla:

Sr. No.	Variety	Year of release	Adaptability
1.	Kufii Sindhuri	1967	Medium maturing, popular in North Indian Plains
2.	Kufri Chandramukhi	1968	Early maturing for North Indian Plains and Plateau region of Peninsular India
3.	Kufri Jyoti	1968	Medium maturing for Southern hills, Early maturing for Plateau of Peninsular India. Suitable for both rabi and kharif seasons.
4.	Kufri Dewa	1973	Late maturing for tarai area of Western U.P Medium maturing, popular in North Indian Plains
5.	Kufri Babar	1980	Medium maturing, popular in North Indian Plains
6.	Kufri Lalima	1982	Medium maturing for North Eastern Plains
7.	Kufri S \vaina	1985	Medium maturing for southern hills
8.	Kufri Megha	1988	Late maturing for Meghalaya
9.	Kufri Ashoka	1996	Early maturing for central and eastern U.P., Bihar and West Bengal, to replace Kufri Chandramukhi
10.	Kufri Jawahar	1996	Early maturing for Punjab, Haryana, M.P., Gujrat, and Kamataka
11.	Kufri Sultej	1996	Medium maturing for Punjab, Haryana, U.P., Bihar and M.P. to replace Kufri Badshah and Kufri Bahar

The varieties of potato commonly grown in Pakistan are: Prima, Alpha, Vekaro, Alladin, Baroka, Cardinal, Desiree, Ultimas, Multa, and Lala Faisal.

Cultivation of Sweet Potato (*Ipomea batatas*)

Botanical Name:

Ipomea batatas.

Family:

Canvolvaceae.

Sweet potato is a important tuber crop grown for starchy food in tropical and sub tropical countries. India it is called as 'Sakharkanda'.

Composition and Uses:

Sweet potato contains starch. Mainly sweet potato is used after boiling, baking and frying. It is important source of starch, glucose, sugar syrup and industrial alcohol. It is the cheapest source of calories. It produces highest food calories among the tuber and root crops. Sweet potato is also used to feed livestock.

Climate:

Sweet potato requires warm weather of at least four to five months. It needs temperature range between 21 to 27 °C. a well distributed rainfall of 75 to 150 cm is desirable. It is a tropical and subtropical crop. It can not tolerate heavy rainfall which induces excessive vegetative tuber development. It does not stand frost.

Soil:

Since the tuber growth takes place inside the soil, it needs loose friable soil for better root development. It can grow in sandy loam and clay sub soil. Heavy clay soil, which becomes hard after drying checks the development of the tubers and highly sandy soils cause development of cylindrical pencil like tuber. In compact soil tuber yield is low due to poor aeration. Such soils also creates harvesting problem. High fertile soils favor luxuriant vegetative growth. Soil should be well drained for successful cultivation. 5.8 to 6.7 soils Ph is suitable.

Planting Season:

Time of planting vine cutting and growing season have marked effect on the yield of sweet potato. In Maharashtra it is planted in September – October. For Kharif crop planting in May- June the cutting from sprouts and higher yield of tubers cutting of 24- 30 cm long are planted. Vines are buried inside the soil in the middle and one node at each and kept exposed.

Methods of Planting:

Sweet potato can be propagated by tubers clips or vine cuttings. Propagation by vine cutting is a very common practice. The cutting are grown in the nursery from the old vines or from clips produced on the tubers planted in the nursery beds. The vine cuttings are planted either on ridges for monsoon crop or in flat bed in low rainfall area. Terminal cuttings from full grown plants are reported to give better growth and yield than the basal or model cuttings. Each cutting should have at least four nodes. The vine cuttings are planted at about 60 cm from row to 30cm within the row. Keeping the end bud free roots is formed in the central part of vines which is put inside the soil. It is better to plant the cuttings after them 8 to 10 minutes in D.D.T 50 % solution.

Seed Rate:

About 40- 50 thousand cuttings are required for planting an hectare of land.

Cultivars/ Varieties:

In Maharashtra white fleshed with red skin colour sweet potatoes are grow. The cultivars are from clonally selection. Variation among the cultivar exists in respect of skin and flesh colour, shape and size of tubers. Mostly local varieties are grown. There are varieties like H- 268, Pusa Safed, Pusa red, Pusa sunahari, NO- 4004, S- 30, Klamath and recent one Skarkand – 5. The period of maturity ranges between 90- 140 days. In general it is ready for harvesting in 120 days.

Sweet Potato:

1. Varsha:

It is released for Maharashtra suitable for rainy season. It is the high yielding (150 q/ ha). it belongs to the group varieties of sweet potato.

2. Konkan Ashwini:

A sweet potato variety Konkan Ashwini is released for cultivation in Maharashtra. It is a short duration, dual seasonal and high yielding variety.

Manures and Fertilizer:

30- 50 cart load FYM should be given at the time of soil preparing. 50- 60 kg N, 50- 60 kg P and 100 – 120 kg k /ha should be applied ½ N, full P and K should be applied at the time of planting. Remaining ½ N should be applied 30 days after planting.

Irrigation:

In Kharif season crop needs less irrigation the most critical stage of moisture supply is 40 days after planting. Excessive moisture supply deteriorates the quality of tubers.

Interculture:

Intercultivation to suppress weed growth is necessary to obtain high yields. Earthing up is followed. Earthing up after 13- 30 days of planting increases the yield. In early of the crop weeding is essential as in later stage vine growth suppresses the weeds.

Harvesting:

Time of harvesting depends on the variety. Before harvest vines are cut and tubers are dug with the help of fork. Sweet potato is harvested when leaves turn yellow and shed. Judge the maturity by cutting tuber. If latex dried up without turning black, it is sure that the sweet potato is matured. Irrigate the crop 4- 6 days prior to harvest to facilitate harvesting. Take care that tubers are not injured while harvesting.

Yield:

Under rain fed conditions yield upto 80- 100 quintal / ha obtained while under good management and irrigated condition yield upto 250 q/ ha is obtained. In an addition to tubers 10 – 25 tonnes of vines are produced per hectare.

Insect Pest of Sweet Potato

1. Sweet Potato Weevil:

Sweet potato weevil is a most serious pest of this crop. A red and blue ant like weevil that makes holes in the stem and tubers. The pale yellow grub of this pest having no feed bores into the vines.

Control Measures:

i) Cultural practices such as destruction of alternate host, removal of infested vines and tubers, deep ploughing of land harvest and earthing up to prevent entry of weevil into tubers have been found effective for controlling for this pest.

ii) This pest can be controlled by spraying the crop with Endosulfan (Thiodane 35 EC) @ 2 ml per liter of water of carbaryl @ 2.5 gm per liter of water.

2. Sweet Potato Sphinx:

It is a serious leaf eating caterpillar of sweet potato. A stout long horn caterpillar that completely that defoliate the plants.

Control Measures:

i) Hand picking and destroying the caterpillar is the best method, if infestation is less.

ii) This pest can be controlled by spraying the crop with lead arsenate of Endosulphan, if infestation is very severe.

Disease of Sweet Potato

1. Leaf Spot:

Large brown spots appear on leaves.

Control Measures:

This disease can be controlled by spraying with Zineb (Dithane Z- 78) or Mancozeb (Dithane M – 45) @ gm pre liter of water at an interval of 15 days.

2. Black Rot:

It is primarily a storage rot of tubers but may also occur in the field, in the field. Infested foliage become yellow and sticky in appearance and black canker are seen on the portion underground and grey black circular spots occur on fleshy roots.

Control Measures:

- i) Careful sorting of tubers before storage and rotation of crops are helpful in reducing the incidence of the disease.
- ii) The seed tubers should be treated with mercuric chloride solution (1 in 1000) or 2.5 per cent borax solution.

3. Soft Rot:

Storage rot of tubers and their tissue become soft and stringy.

Control Measures:

- i) Avoiding brushing and injuring the tubers at harvesting is the best method of controlling this disease.
- ii) The walls and floors of storage room should be disinfected before the storing tubers.

Cultivation of Okra or Lady's Finger or Bhendi (*Abelmoschus esculentus*)

Botanical Name:

Abelmoschus esculentus L. Syn. *Hibiscus esculentus*

Family:

Malvaceae.

Bhendi is one of the popular vegetable in India. It is cultivated extensively all the year round for its immature fruits. Bhendi fruits are used as a vegetable. The stem of the plant is used for the extraction of the fiber.

Climate:

Bhendi is a warm season vegetable crop and requires a long warm growing season. In India it is grown in summer months and during the rainy season. It is highly sensitive to frost. Seeds germinate poorly at ground temperature of 20 °C or less.

Soil and its Preparation:

Bhendi can do well in all kinds of soil. .But sandy loam and clay-loam soils are best for its cultivation. The optimum pH range is between 6 and 6.8. The land is prepared by giving 5-6 ploughings. Cartloads of FYM or Compost should be incorporated during land preparation.

Seed Rate & Sowing Time:

For kharif season seeds are sown in the month of June - July. The seed rate for rainy season is 8 to 10 kg. For summer season it is sown in the month of January-February. The seed rate for summer season is 12 to 15 kg/ha. For winter season it is sown in August"- September.

Layout & Spacing:

Ridges & furrow type of layout is use. Sometime bhendi is also sown at the border of main crop & layout is same as for main crop. For summer crop spacing is 45x30 cm. & Rainy season crop spacing is 60 x 45 cm.

Improved Varieties:

Pusa Makhamali, Pusa sawani, Vaishali, Vagmi, pb. Padmini, Selection 2-2, Pusa sawani is fairly mosaic tolerant Parbhani Kranti (Resistant to virus disease). Okra (*Abelmoschus esculentus*).

1. Parbhani Kranti:

It is yellow vein mosaic resistant variety released in 1985 by MAU, Parbhani- It is derived from back cross of *Abelmoschus manihot* x *A. esculentus*. Two back crosses were made using Pusa sawani (*A. esculentus*) as a recurrent parent followed by selection in the subsequent generations- It is a yellow vein mosaic virus (YVMv) resistant variety isolated in F8 generation. Plants flower in 45-60 days. Fruits are dark green in colour, smooth, lender and slender.(MAU, Parbhani)

2. Phule Kirti:

It is the F1 hybrid of okra suitable for both the kharif and summer seasons. It is also suitable for warm and moderate humid regions. Stem is green in colour with red patches. Fruits are green in colour. It is resistant to yellow vein mosaic virus. Shelf life is 3-4 days. It gives on an average yield of 163 quintals per hectare. Besides it is released through the State Variety Release Committee of Maharashtra, it is also notified by the Central Sub Committee on Crop Standards, Notification and Varietal Release at the national level.(MPKV, Ramrri)

Manures & Fertilizers:

20 to 25 tonnes well rotted FYM mixed with soil during field preparation. For rainy season, crop apply 50 kg; N, 50kg P2O5 per hectare. For summer season 40 kg N, 40 kg P2O5 should be applied.

Interculture Operational:

Weeding, thinning and earthing up is the important intercultural operations of bhendi. Earthing up in the rows should be done in rainy season crop. Application of herbicides for controlling weeds in bhendi is found effective. Basalin (Fluchoralin 48 per cent) @ 1-2 liter per hectare, and soil application of Tok-E-25 @ 5 liter per hectare effectively controlled the weeds.

Irrigation:

Pre sowing irrigation should be given in the soils having no sufficient moisture. First irrigation is given just after sowing and then the field is irrigated after 4 to 5 days in summer and 10 to 12 days in winter season.

Harvesting:

Flowering begins from 35 to 40 days after sowing and fruits are ready for harvest four to five days after flowering. Bhendi pods may be harvested continuously at some intervals. The young fruits can be harvested in the morning. Delay in harvesting may make the fruits fibrous and they lose their tenderness and taste.

Yield & Crop Duration:

Rainy season crop gives 7500 kg/ha. Summer crop gives 5000 kg/ha. Duration is 100 & 90 days respectively.

Insect Pests of Bhendi

1) Stem and Fruit Borer:

The caterpillar bores into the growing points of the plant and later on it also attacks flower bud and young fruit.

Control Measures:

Spraying the crop with carbaryl (Sevin-50 W, Sevimol 0.1 per cent, or Monocrotophos (Nuvacron) 0.1 per cent or Endosulfan (Thiodan -35 EC) 0.07 percent on the 40 and 60 days after sowing is found beneficial.

2) Jassids:

The hoppers suck the cell sap from the leaves, causing curling of leaves.

Control Measures:

Spraying the crop with Diclorovos (Nuvall) 76 EC or Malathion -50 EC @ 2 ml per liter of water or Dimethoate (Roger -30 EC) or Oxydemeton Methyl (Metasystox -25 EC) @ 1 ml per liter of water can control this pest.

Cultivation of Fenugreek or Methi (*Trigonellia foenum-graecum*)

Botanical Name:

Trigonellia foenum-graecum L.

Family:

Leguminosae.

Local Name:

Methi- Hindi. Venthayam- Malayalam.

Fenugreek is used as vegetables as well as pulse. The leaves and young pods are used as vegetable and the seed as condiments. It has also some medicinal value. It prevents constipation removes indigestion stimulates the spleen and is appetizing and diuretic. The leaves are quite rich in protein minerals and Vitamin C.

Climate:

Fenugreek is cool season crop. It is fairly tolerant to frost. Common methi is growing and produces upright shoots whereas kasurimethi is slow growing and produces upright shoots whereas Kasurimethi is slow growing initially and remains in a rosette condition during most of its vegetative periods.

Soil and its Preparation:

Fenugreek can be grown on any type of soil, but loamy soils are best suited for its cultivation. The soil should be free bad drainage. The land should be prepared thoroughly.

Seed and Sowing:

For leafy vegetables methi is sown from middle of September to middle of March. The seed rate per hectare is 25 kg. Soil is thoroughly prepared before sowing. Flat beds of convenient size should be prepared. Seeds are sown by broadcasting but line sowing at a distance of 25 cm. facilitates weeding and intercultural operation.

Manuring:

30 to 50 cart loads FYM should be applied at the time of soil preparation. 50 kg N/ha should be applied. ½ dose of N should be given at the time of sowing. Remaining ½ doses is again divided in low equal dose which are given at the time of first and second harvesting.

After Care:

One or two weeding may be given during early stages of growth. Soil should contain sufficient moisture. The first irrigation is given just after sowing. And subsequent irrigation are given are given at seven to ten days interval.

Harvesting and Yield:

Normally common methi leaves are nipped 2 cm above ground after 3- 4 weeks of sowing. In general 2-3 cutting are taken. When plant is uprooted crop is boundled and marketed.

Varieties:

Kasuri, pusa, Early, punching type, Methi No- 47, No- 14, EC- 4911 are the important varieties of methi.

Pusa early bunching is early cultivar, it is a common methi. It takes about 125 days for its seed maturity. Methi No.47 and Methi No.14 are high yielding varieties released by Maharashtra Department of Agriculture.

Cultivation of Coriander (Corandrum sativum)

Botanical Name:

Corandrum sativum L.

Local Name:

Dhane- Marathi, Dhaniya- Hindi, Kothamlli- Malayanum.

Family:

Umbellifera.

Coriander is indispensable spice in the kitchen. It gives good flavour to dish. Thus coriander seed and fresh leaves are common used in every kitchen to prepare tasty dishes. Coriander leaves constitute one of the richest source of Vitamin 'C' and condiment in curries and in Fresh chutneys.

Climate:

It is a cool season crop. Cool weather is favorable for its cultivation. It is susceptible to mild frost.

Soil and its Preparation:

It grows well on all type of soil. But sandy loam and clay soil are best suited for it. The land should be prepared thoroughly.

Sowing:

Seeds of cultivar are crushed and then sown. It is generally broadcast. Seeds be soaked in water for overnight to remove inhibitor and thereby enhance germination. About 40 kg seed is sufficient for sowing an area of one hectare if grown for its leaves. Sowing of crop at an fixed interval be carried out for regular marketing. It is excellent in tomato, brinjal, cabbage and cauliflower.

Fertilizer:

Besides FYM, apply 90kg Nitrogen, 80 kg Phosphorus and 50 kg Potash per hectare. Adopt weeding as required.

Harvesting:

When grown for seed purpose, it is harvested. After full development of grain. If grown for its tender fresh leaves. It is harvested before flowering. Harvest, when it attains a height of about 20 cm. uproot the plants and tie bundles of convenient size for marketing. One can harvest 10 tonnes of coriander from an hectare.

Varieties:

Following are the improved varieties of coriander: T- Np (D) -92, NP (D)-95, NP (J)-16, NP (J) – 21 4M, Delhi local, strain- 85 etc.

Cultivation of Amaranths (Amaranthus tricolor)

Botanical Name:

Amaranthus tricolor L.

Family:

Amaranthaceae.

Local Name:

Math- Marathi, Chhoti Chulai- Hindi, Cheera- Malayalam.

1. *Amaranthus candatus*: rajgira.
2. *Amaranthus tricolor*: Math.
3. *Amaranthus blitum*: tamda math.
4. *Amaranthus oleracea*: tandulja.

It is a warm season leafy vegetable grown in Kharif and summer season. It is short duration crop. Leafy types are grown in kitchen and market garden, recently grain types have gained some importance in protein, minerals and vitamin A and C.

Soil and its Preparations:

It can be grown on all types of soil, but loam soil is most suitable for its cultivation. Fine seed bed necessary for this crop and this can be done by repeated ploughing.

Sowing of Seed:

The crop is suitable for growing in summer as well as rainy season. The seed is sown in December-January for summer crop and April – May for rainy season crop. Seeds of this crop are very small. The seed is sown broadcast in sown or in flat beds after mixing those with fine soil for well distribution. The seedlings of badi Chauli are transplanted after raising them in the nursery during rainy season at a spacing of 45 cm from row to row 30 cm. from plant to plant. About one kilogram seed is required for sowing one acre.

Manuring:

The plants are top-dressed with Urea @ 25 kg/ ha. After 15-20 days of sowing and subsequent top dressing are done. After each cutting.

After Care:

The crop becomes ready for harvesting about three to four weeks after sowing and subsequent cuttings may be done after 7 to 10 days. The plants are harvested by cutting them periodically. This crop regrows quickly after each cutting.

Yield:

The yield of green leaves is 90 to 100 q/ha and seed is 200 kg/ ha.

Varieties:

The recommended varieties are badi Chauli, Chotti Chauli and Co- 1. *Amaranthus*: (*Amaranthus* sp.)

1. Konkan Durangi:

This is developed by selection. It is high yielding giving 82 % more yield than control. The upper surface of leaves is green while the lower one is red. Pedicel and stem colour is also red. It is suitable for both the rabi and summer seasons.

Insect Pests Management in Green Vegetables

1. Aphids:

They are very small insect pests found generally in groups on tender shoots and leaves of plant. They suck up the cell sap and make them poor in growth.

Control Measures:

- i) Spray 40 % commercial Nicotine Sulphate (1 to 800 parts of water) about 80 – 100 gallons will needed of this chemical for spraying in an hectare.
- ii) Spray Pyrethrum 80 – 100 gallons per hectare (after diluting 1 part of it in 1000 parts of water).

2. The Painted Bug:

The painted bug and flea beetle attack scars on plants the former is a sucking insect while the latter makes holes in leaves. The bug can be controlled by 5 % or 0.5 %, rotenone dust. The beetle succumbs to a spray of lead Arsenate (½ kg /500 liter of water) or 0.25 % DDT spray.

3. Laphygma Exigua:

Its caterpillar attack is a host range covering almost all leafy vegetables, and the caterpillar of *Plusia orichalora* attacks fenugreek, Palak and a few other kinds.

4. Brassica:

The Brassica are subjected to the attack of large larvae of mustard sawfly, particularly early in the season. The larvae feed on leaves by cutting small holes and ultimately skeletons the plants.

Control Measures:

- i) Pick the larvae of sawfly and destroy it.
- ii) Spray with Pyrethrum extract 1: 800.
- iii) For the caterpillar of the pests described above 0.2 to 0.3 % Pyrethrum dust has been reported to be effective.

Diseases Management in Green Vegetables

1. Leaf Spot Diseases:

It is caused by *Cercospora traversina* in fenugreek, *C. baticola* in Palak *C. Spinacea* in spinach, *C. Chenopodi* in bathua and *Cercospora* sp. In poi.

2. Downey Mildew Disease:

It is caused by *Peronospora trigonella* on fenugreek, *P. parastica* on brassicas, *P. effuse* on *Chenopodium*.

3. White Rust:

In Brassica it is caused by *Albugo vandila*, *A. blightii* in *Amaranthus* and *A. portulacae* on Purslane. Brassica leaves stem and pods are spoiled by lesions:
It is caused by *Alternaria Brassicae*.

Control Measures:

Removal of the infected plants and their destruction would be helpful reducing losses from these diseases.

Salad Crops

Salads are vegetables which are usually eaten uncooked. They are appetizing and have high mineral and vitamin content. These are seasonal crops.

1. Lettuce

2. Celery

Cultivation of Lettuce (*Lactuca sativa*)

Botanical Name:

Lactuca sativa L.

Family:

Compositae.

Among the salad crops, the garden lettuce is the most popular vegetable grown on a commercial scale. This is a tall annual leafy herb with a milky juice. It produces a short sitcom early in the seem, cluster of leaves varying considerably in shape, character and colour in different varieties. Later in the season a seed stalk is produced. It thrives best during the coller parts of the growing season and to be of the highest quality. It is almost entirely self pollinated crop.

Uses and Nutritive Value:

Lettuce is a nutritious leafy vegetable, rich in mineral and a source of vitamin. It is now considered as the most important salad crop and used particularly as the base for salad. It is anodyne, sedative, diuretic and expectorant. Its seeds are cooling, demulcent and refrignant, while its leaves are slightly lypnotic and sedative. The composition of lettuce is as follows:

1. Edible Portion: 66 %
2. Moisture: 93.4%
3. Protein: 21 g
4. Fat: 0.3 g
5. Mineral: 1.2 g
6. Fiber: 0.5 g
7. Other carbohydrate: 2.5 g
8. Calories: 21

Minerals:

1. Calcium: 329 mg

2. Total Phosphorus: 80 mg
3. Total iron: 2.6 mg
4. Riboflavin: 0.13 mg
5. Nicotinic acid: 0.5 mg and
6. Chlorine: 178 mg.

History and Origin:

Lettuce has been cultivated for more than 2500 years. Cultivated lettuce originated around Asia Minor, Iran and Turkistan. Lettuce is an annual crop and belongs to the family Compositae.

Climate:

Lettuce does well in a relatively cool growing season with a monthly average temp, of 12.8 to 15.6 C. higher temperature induce bolting and cause bitter taste in the leaves and accelerate the disorder 'tipburn' and rot. Seed germination is also affected at temperature above 27 °C.

Soil:

Lettuce will tolerate a variety of soil but performs best when grown on a rich friable and well drained soil. Soils rich in organic matters have a high water holding capacity and ensure uniform moisture supply. These conditions are best suited for it, as it has a very shallow root system. Lettuce is sensitive to high acidity. The best soil reaction is pH 5.8 to 6.6. the yield is low at pH 5 or above 7.

Manuring:

The lettuce plants have mostly surface roots. The surface soil, therefore, should be well supplied with nutrients. The fertilizer requirements will vary with the soil. Usually 10-15 tones of FYM, 85 kg N, and 60 kg each of Potash and Phosphorus per ha when give good results.

Sowing:

The main sowings are done after rains. i.e early October to November. Leaf types are sown either directly on moist soil or started in a nursery and later transplanted. Since the seeds are tiny, the land is prepared thoroughly and seeds are sown in rows 15 to 25 cm apart. Dropping the seeds very close together and 1 cm deep and later thinned to 3 cm and when these begin to crowd alternate plants are removed for use. Seed rate is 2 ½ kg per hectare.

The nursery sowing is done. In September- October and transplanted when the seedlings are 5 weeks old. They are planted 20 to 25 cm apart in rows which are 30 to 40 cm apart. About 0.5 kg of seed required for raising seedlings for one hectare.

Germination is affected by temperature. Lettuce seed germination at a range of temperature but the optimum is 18- 21 °C.

After Care:

Hoeing and weeding are intermittently done for proper aeration and keeping down the weeds. The cultivation should be very shallow since the crop has root system near the surface of the soil.

Water should be supplied to keep uniform moisture condition in the soil. A frequency of 4 to 7 days may be enough for each irrigation. Drainage is as important as supplying irrigation, for excess of soil moisture may cause rooting.

Harvesting:

Leaf varieties are harvested when the leaves are tender and immature. First harvesting is done 35 days after sown. In case of head varieties the crop is harvested when it has attained a good size and solid head. While handling care should be taken not to damage the wrapper leaves.

Yield:

The yield per ha is 10,000 to 12,000 kg in case of head lettuce. Open leaf types generally give comparatively higher yields.

Storage:

Lettuce can be kept for 2- 3 weeks at 0 C and 90- 95 % RH.

Seed Production Technique:

Lettuce is mainly self pollinated crop. There is on there is no difficulty in the production of seeds at the leafy varieties. All the seeds do not ripe at the same time in it, so they should be collected as they ripe. The seed harvesting is completed by the end of May.

The leafy types produce about 500- 600 kg whereas, the variety great lakes produce only about 100 to 125 kg per ha.

Varieties of Lettuce

All the varieties many be divided into following classes which differ greatly in their texture, from and colour.

1. Loose leaf Varieties:

It is spreading in habit with leaves arranged in a cluster or a bunch. There is no over- lapping of the leaves. They are quick growing varieties. Ex. Grand raphids, Black seeded Simpson, prime head.

2. Crisp Head Varieties:

They are of cabbage heading habit. In them the central leaves are much over klapped. The foliage is crisp or brittle. The varieties of this class can not be used for a continuous supply of leaves during the season. Ex. Lmperial, Empire green land.

3. Butter Head Varieties:

They are like crisp bvarieties but with comparatively soomothy, finely veined leaves of soft texture of soft texture and buttery flavor and oily touch. Ex. White Boston, May King, Bibb, Midas.

4. Cos Lettuce or Romanic Variety:

It is an up right variety with long head and supatulatc leaves. It is much in demand due to its brittle and crisp leaves of high quality. Ex. Dark green Cos, parris island Cos. Valamine.

The recommended varieties from the I.A.R.I, New Delhi are great lakes, Imperial 847, Chinese yellow and slobolt, may king, Paris rohitc.

Insect and Disease Management in Lettuce

Insect Pests:

Aphids are the most troublesome insect on lettuce. The use of 3 or 4 per cent Nicotine dust or spraying of 40 % commercial Nicotine sulphate gives satisfactory control if applied when the temperature is around 18 to 21 °C.

Diseases:

1. Slimy Soft Rot:

It is the most important disease particularly in head lettuce. It can be controlled by early removal of the heads and also by keeping the soil uniform in moisture and keeping the surface soil relatively dry.

2. Downy Mildew:

This disease is caused by *Bombya lactucae*. The typical symptoms are yellow angular areas, delimited by young veins, appearing on the upper leaf surface and sporulation on the lower surface. The lesions turn brown and may coalesce.

Control of disease is by use of resistant varieties like Imperial 17, Calmar systemic fungicide effectively control this disease.

3. Mosaic:

It is a virus disease first noticed in the seedling stage. There is a slight inward rolling of the leaves along the main axis followed by a light green to yellow mottling of leaves. There may be severe stunting and a dull green to yellow mottling leaves. There may be severe stunting and a dull green to slightly yellow discoloration of the whole plant. The disease is seed borne and use of disease free seeds is the only control measure.

4. Tip Burn:

It has been a difficult problem because of multiplicity of causes calcium nutrition seems to be basic leading to tip burn. Less calcium is found in tip burn of lettuce. Lenticels rupture near the leaf margins, releasing latex into the surrounding tissue. This causes dark brown spots to form at the points of rupture. Most effective methods of control are to cultivate resistant varieties such as Salinas, Calmar and Monte mar.

Cultivation of Celery (*Apium graveolens*)

Botanical Name:

Apium graveolens var. *dulace*.

Family:

Umbelliferae.

Celery ranks second in importance amongst salad crops. In India this crop is not commercially as a vegetable crop. Celery as a salad crop is mostly grown in kitchen or home gardens. In colder climate and on the hills. Celery is a biennial plant, while at plains it becomes an annual. It has dark green foliage with less developed stalks and swollen roots.

Uses and Nutritive Value:

The leaf stalks or petioles of celery are eaten as salad, in soups, in puree, fried and spiced. The composition of edible portion is as follows: Moisture 93.5 gm, Phosphorus 38 mg, Vitamin A 867 L.U, Protein 0.8 g, Minerals 0.9 gm, Calcium 30 mg, Iron 4.8 mg. Fiber 1.2 mg, Vitamin C 62 mg and calories 18.

History and Origin:

The place of origin of celery extends from Sweden Southward to Algeria. Egypt, Abyssinia and in Asia even to the Caucasus, Baluchistan and the mountains of India. It belongs to the family Umbelliferae. In India quite a large area in the Punjab and U. P is devoted to the production of the seed. There are annuals and the seeds are being exported to European countries as a condiment.

Varieties:

There are green and yellow or self blanching varieties. Giant Pascal, Emperor of Jean and Golden Self blanching are popular varieties. The varieties recommended by I.A.R.I are standard bearer and Wright Glove Grant.

Climate:

It is a moisture loving plant requires a long, cool seasons with a well distributed moderate rainfall during the growing season. It does not stand severe freezing climate. Plenty of sunshine during day and cool nights is beneficial.

Soil:

Muck or peat soils are commonly used due to their high water holding capacity and friability. Alternatively any rich, friable, well drained and deep soil is good and is most suited to clayey soils.

It does not thrive well on a soil more acidic pH 5.5. It does not perform well at a pH above 6.7.

Manuring:

Celery is a shallow rooted crop. So the majority of nutrients are drawn from the upper surface. Good quality depends on continuous steady growth. The nutrient requirements of celery are quite high. It requires a carefully prepared and heavily manured about 15 tonnes FYM per hectare, during preparation. Fertilizers upto one tonne per hectare are applied once after transplanting and establishment and one to two dressings are given at intervals of three weeks.

Sowing Seeds and Transplanting:

The seeds of celery which are really fruits take a long time for germination and require to be sown in well prepared beds. The seeds are sown from July to September in the nursery bed and need a partially shady and cool situation. The seeds are sometimes kept moist for a few days before sowing to help in germination. About 300 to 400 gm of seeds are required for one hectare. The seeds are lightly covered and seedlings come out 3 to 8 weeks and are thinned to about 1.2 cm spacing. The seedlings are transplanted after about two months to a spacing of 45 x 15 or 60 x 15 cm. the planting should not be done very deep as spacing damage may result.

Irrigation:

The supply of water must be ample all times especially at planting, amnuring and blanching.

Blanching:

Excluding light from the stalks while the plants are still growing makes them devoid of chlorophyll and are known as blanched. The plants are earthed up when they are about 40 cm tall, after removing suckers. The earth should not be allowed to come into contact with the heart of the plant. Two more earthlings are allowed at fortnightly intervals and only tips of leaves should be visible after the final earthling up. Blanching improves both the flavours and the tenderness.

Weeding:

Frequently weeding is done. Care should be taken twit the roots are not injured by implements, as they are shallow.

Harvesting:

The harvest is taken up four to five months after sowing. Each plant is cut just below the surface with a sharp knife. The plants are then trimmed and prepared for the market.

Yield:

The average yield of about 25000 kg/ha.

Storage:

Celery in storage absorbs foreign flavours. So it should be kept away from the odours of other products. They are stored after harvest in trenches, cellars and cold storage. In cold storage they are kept at 0 C at 95 – 98 % RH.

Insect and Pest and Disease:

Celery leaf miner, carrot weevil and Aphids are come of the insect which attack celery.

The disease, which affect celery are Early and late blight. Fusarium, yellow and pink rot. The virus diseases affecting celery are celery mosaic and Aster yellow.

Perennial Vegetables

Perennial vegetables are those which produce the protein continuously for several years. They occupy the land for a number of years, and therefore, should be located where they will not interfere with normal tillage operations for other crops. These crops have tuberous roots, rhizomes or crowns.

The upper portion may dry up during very cold winter months and again sprout with the coming of spring. The vegetables which grouped as perennials are Asparagus, Rhubarb, Artichoke and sea kale. A number of cucurbits like cho-cho can also be considered as perennial vegetables. There are some fruit crops too which are commonly cooked as vegetables in the immature stage. They are Plantain, Papaya and jack Fruit.

Cultivation of Asparagus (*Asparagus officinallis*)

Botanical Name:

Asparagus officinalis L.

The asparagus is cultivated for its tender shoots, commonly known as spears. It starts yielding a sizable crop after about three years and with good care gives an economic yield for about 10-15 years. The yield goes on increasing for 6-7 years, then remains uniform up to about 12 years, after which it gradually declines. The best quality is obtained from the fourth to tenth year.

Uses and Nutritive Value:

In addition to its use as a fresh vegetable, asparagus is also preserved in cans or bottles. A white crystalline substance "Asparagins" freshly cut and immediately served is one of the most delicate, wholesome and appetizing products of the home garden. The spears are treated as a delicacy in preparation of soups and white crystalline substance "Asparagins" $C_4H_5O_3N_2H_2O$ is obtained from the juice of young shoots. It is considered to be a good diuretic and is used especially in chronic dropsy and chronic gout. It was known to the Romans as a medicinal plant and then as a vegetable.

The edible portion contains (%) Moisture 93, fat 0.2, Ca 0.2, Fiber 0.7, Starch 0.4, Protein 2.2, Ash 0.6, total CHO 3.9, Sugar 1, 2.

History and Origin:

It was used by the Romans centuries before the Christian era. It is indigenous to Europe and Asia. It belongs to the family Lilaceae and it is a perennial dioecious herb growing 1 to 3 m height.

Varieties:

There are two general types of asparagus based on the colour of the spears.

i) The less important group produces light green or whitish spears. Ex. Mammoth white and Conserve Colossal.

ii) The most important group includes purple spears. Ex. Marv Washington. The tips of the spears before exposure to the light are purplish and the spears become dark green in sunlight.

The recommended variety of IARI New Delhi is Perfection.

Climate:

Though well adapted to temperate regions and in the hills in tropics and subtropics, it can be grown in the plains also.

Soil:

A rich, friable, well-drained soil is required. A high fertility, plentiful supplies of organic matter, good drainage and an abundant supply of moisture are factors favorable to the crop. The optimum pH requirement is between 6.0 and 6.7. Proper preparation of soil is essential as the crop occupies the soil for a number of years.

Sowing:

In the plains July- November is the common period for sowing. The seeds are sown in a well-prepared nursery bed in rows 15 to 22 cm apart and 1-2 cm away from each other and at a depth of 2.3 cm. They are thinned to a spacing of 5 cm. The seedlings are allowed to grow for one year in

the nursery bed and are then planted at the permanent site. Seeds take 3 to 4 weeks to germinate. About 2 ½ to 4 kg seed will be required for a hectare.

Planting:

The age of transplanting is one year seedling or crown. These plants which have a good growth and branching should be selected. The crows are dug up from the nursery at a depth about 20cm. the rows are spaced at about 1.5 to 2 m and 45 to 60 cm distance is kept within the rows.

Cultivation of Palak and Vilayatipalak (Beta vulgaris)

Botanical Name:

Beta vulgaris L, and

Family:

Chenopodiaceae.

Local Name:

Palak: Marathi, visalacheera: Malayalam.

Amongst all the vegetables, the leafy vegetables have a very high protective food value. They are rich in mineral and hence can be called as "Mines of minerals". Vitamin A and C are present in abundant quantities. Beside this, soft fibrous matter is specially in providing necessary roughage in diet.

It is a widely grown leafy vegetable. It is rich and cheap source of vitamin A, iron, essential amino acids Ascorbic acid etc.

Climate:

Spinach beet is a hot weather crop but can also be grown during moderate winter season. It can also tolerate frost to some extent. Spinach does well during winter season. It is extremely susceptible to water logging and bad drainage. It can be grown successfully in plains and on hills.

Soil and its Preparation:

Though it can be grown on any kind of soil that is free from bad drainage yet sandy loam and alluvial soil are best suited for them. Their growth is checked in acidic soil. They need a Ph between 6.0 to 7.0 for optimum growth. The soil is prepared by ploughing 4 to 6 times and then leveling by planking. The beds and irrigated channel are made sowing it in the field.

Sowing:

It is sown from October as main crop and in January- February as summer crop in this region. It requires 30 -40 kg seed per hectare. Seeds are soaked in water before sowing. Seeds are generally broadcast in flat beds. However, line sowing at 20 cm distance is convenient for intercultural operation. Follow light irrigation after sowing.

Manures and Fertilizers:

30 to 50 cart loads FYM should be at the time of soil preparation. 50 kg N/ha should be applied, ½ dose of N should be given at the time of sowing, remaining ½ dose is again divided in two equal doses which are given at the time of first and second harvesting.

Interculture Operation:

One to two weeding should be given to keep the crop clean and healthy.

Irrigation:

Winter crop needs irrigation 10 -15 days interval depending on soil type. January- planted crop needs irrigation at 6 -7 days interval in summer months.

Harvesting:

The crop becomes ready for first cutting after 35 – 40 days of sowing. Only well grown succulent and tender leaves be trimmed. Winter crop gives more cuttings. On an average Palak crops gives 4 -6 cuttings. The total yield of green leaves is 80- 100 quintals per hectare. After harvesting, leaves are tied in bundles and sent for marketing.

Varieties:

1. Pusa All green:

It produces uniformly green tender leaves and gives about 6 cuttings depending upon time of sowing and management. It is heavy yielder giving leaf yield of 125 quintals per ha.

2. Pusa Jyoti:

It produces large, green, thick, tender, succulent and crisp leaves of good taste. Plants are vigorous quick growing and regenerate quickly after each cutting. It gives 6 – 8 cuttings and yield over 200 – 400 quintals per hectare. It performs well in this region.

Other Varieties:

Besides, there are other varieties viz. Pusa harit and jobner Green.

Present Status of Fruit Growing

The country well as Maharashtra possess a diversified soil and climatic condition and holds a pride place with regard to production of tropical to temperate fruit with more or less perfection. Southern states having tropical climate grow successfully many tropical fruits like coconut , areca nut, cashewnut, spices, among while northern states like Kashmir , himachal Pradesh, northern Paris of Punjab and U.P, kulu valley, kumao hills and Nilgiri and Palni hills in south having temperate climate grown successfully many temperate fruits like apple, pear, peaches, plumes etc.

A variety of such tropical fruits are grown in Maharashtra. The area under fruits in Maharashtra in 1986- 87 was 115000 ha having total production of 1901450 tons of 16.48 tons/ ha the area under fruits in Maharashtra of 1989-90 is as under.

Area Under Fruits in Maharashtra of 1989-90:

Sr. No	Name of fruit	Area(ha)
01	Banana	53087
02	Mandarin orange	36888
03	Mango	23032
04	Cashew nut	14448
05	Grapes	14079
06	Kagzi lime	6940

07	Coconut	6625
08	Pomegranate	6424
09	Sweet orange	5677
10	Guava	4511
11	Sapota	2424
12	Areca nut	1818
13	Custard apple	1745
14	Ber	1656
15	Papaya	1512
16	Fig	333
17	Other fruit	2300
Total	=	184129Ha